CFA® LEVEL III · PORTFOLIO MANAGEMENT PATHWAY · FOR THE FEBRUARY 2027 EXAM

CFA L3 Portfolio Management Pathway Summary

Every reading (P1–P8) of the Schweser 2025 Portfolio Management pathway notes, explained from zero — 115 subtopics. The pathway is 30–35% of the Level III exam (about 7 of the 20 scored question sets). Each subtopic opens with 'In plain words', then tables, formulas, worked examples and exam traps. Stars combine exam weight, essay likelihood and depth in the curriculum.

★★★★★ must know, asked often ★★★★☆ high ★★★☆☆ medium ★★☆☆☆ low ★☆☆☆☆ skim

‡ rule or number that changes often; check workbook edition

Ch	Chapter	Marks	Priority	Subtopics	5★
0	Exam Trend & How to Use This Summary	—	★★★★★	3	2
1	P1 Index-Based Equity Strategies	4% of PM	★★★☆☆	8	4
2	P2 Active Equity Investing: Strategies	12% of PM	★★★★☆	17	14
3	P3 Active Equity Investing: Portfolio Construction	11% of PM	★★★★☆	19	12
4	P4 Liability-Driven & Index-Based Fixed Income	14% of PM	★★★★★	18	13
5	P5 Yield Curve Strategies	13% of PM	★★★★★	16	12
6	P6 Fixed-Income Active Management: Credit Strategies	19% of PM	★★★★★	12	8
7	P7 Trade Strategy and Execution	12% of PM	★★★★☆	8	7
8	P8 Case Study: Institutional (University Endowment)	15% of PM	★★★★★	10	9
F	Formula Sheet (all readings)	revise last	★★★★★	4	4

5★ list (85) — read these first

Ch 0 · Level III exam structure (Feb 2027) · Where the pathway marks come from

Ch 1 · Three types of factor-based strategy · Three ways to get index exposure · Building the portfolio: replication, sampling, optimisation · Tracking error: causes and control

Ch 2 · Fundamental vs quantitative approaches · Bottom-up vs top-down · Bottom-up value sub-styles · Bottom-up growth sub-styles & worked selection example · Factor-based strategies: rewarded factors & hedged portfolio · Activist strategies · Statistical arbitrage & market microstructure · Event-driven strategies & risk arbitrage · Creating a fundamental strategy & its pitfalls · Creating a quantitative strategy: steps & data · Backtesting: factor scores and information coefficient (IC) · Pitfalls in quantitative investing · Style classification: holdings-based (Morningstar style box) · Returns-based style analysis, self-identification, pros & cons

Ch 3 · Active return and its three sources · The building blocks of portfolio construction · Fundamental law of active management · Active Share · Active risk & how it differs from Active Share · Classifying managers with Active Share and active risk · Absolute risk: contribution of an asset and of a factor · Heuristic vs formal risk constraints · AUM, liquidity, turnover and market impact · Long-only vs long/short: factors in the choice · Long/short structures: gross, net, 130/30, market-neutral · Long/short: benefits and drawbacks

Ch 4 · Duration refresher: Macaulay, modified, price vs reinvestment risk · Effects of yield curve changes on a barbell immunising a bullet · Rules for immunising a single liability; money duration · Multiple liabilities: cash-flow matching, duration matching, contingent immunisation · Closing a duration gap with futures (derivatives overlay) · Defined benefit pensions: liabilities and a futures hedge · Closing the gap with interest rate swaps · Swaptions for the duration gap (worked) · Choosing the optimal swap hedge (rate scenarios) · Risks of liability-driven investing · Bond indexing: why it is harder than equity · Matching the primary risk factors · Passive bond exposure: other vehicles

Ch 5 · The yield curve and how it changes · Five-step return decomposition; convexity; bullet vs barbell · Static yield curve: strategies when your view = the market's · Worked: bond vs receive-fixed swap over six months · Divergent view on level · Divergent view on slope · Divergent view on curvature: butterfly trades · Key rate durations: reading a manager's view · Across currencies: unhedged foreign bonds · Hedging: covered vs uncovered interest parity · Unhedged cross-currency: the carry trade · Evaluating a yield curve strategy (scenario analysis)

Ch 6 · Credit risk, spreads and the credit cycle · Fixed-rate spread measures · Spread duration, DTS and excess spread · Bottom-up credit: models & relative value · Liquidity risk & tail risk (VaR) · Credit default swaps: pricing · CDS long/short and curve trades · Static & dynamic credit curve strategies

Ch 7 · Why managers trade · Inputs to trading strategy & the trader's dilemma · Reference price benchmarks · Execution methods & algorithms · Implementation shortfall (worked) · Evaluating execution: arrival, market-adjusted, added value · Trade governance & best execution

Ch 8 · How to read this case · Managing liquidity risk: four tools · QU endowment: facts you need · Proposed allocation: justify it & name the trade-offs · Why liquidity gets worse in a crisis · Ethics in manager selection (who broke what) · Equity exposure: ETF vs futures vs total return swap · Rebalancing: cash market or derivatives? · ESG policy & the child-labour case

Ch F · Equity (P1–P3) · Liability-driven & yield curve (P4–P5) · Credit (P6) · Trading (P7)

4★ list (24)

Ch 1 · Single-factor vs multifactor models (Fama-French five) · Factor-based strategies: passive or active? · Sources of return and risk

Ch 2 · Top-down strategies & smart beta · Factor timing (style rotation) worked example

Ch 3 · Approaches to portfolio construction · Objectives and constraints · Risk budgeting: the four steps · Relative (active) risk contribution · Choosing the right level of risk · The well-constructed portfolio

Ch 4 · Portfolio statistics from aggregate cash flows · Contingent immunisation in practice · Enhanced indexing: stratified sampling, ESG and adding value · Selecting a bond benchmark

Ch 5 · Volatility views: embedded options · Volatility views: stand-alone options · Hedging foreign coupon bonds: fixed-fixed cross-currency swap

Ch 6 · Floating-rate note spreads · Top-down, factor and ESG credit strategies · Global credit, emerging markets · Structured credit & fixed-income analytics

Ch 7 · Trading by market

Ch 8 · The illiquidity premium

Source: Kaplan Schweser 2025 CFA Level III Portfolio Management SchweserNotes (readings P1–P8 and formula sheet). Weight estimates: CFA Institute's 30–35% pathway share, split by curriculum depth. The core readings (asset allocation, portfolio construction, performance, derivatives & risk, ethics) are not in this file. CFA® is a trademark of CFA Institute.

00 Exam Trend & How to Use This Summary

≈— marks · 3 subtopics · chapter priority ★★★★★

Level III exam structure (Feb 2027) ★★★★★

Item	Detail
Question sets	11 item sets + 11 essay (constructed response) sets, 12 points each; 20 scored, 2 trial
Item set	Short vignette + 4 multiple-choice questions
Essay set	Vignette + questions you answer in writing: justify, calculate, 'determine and justify'
Core share	≈ 65–70% (asset allocation, portfolio construction, performance, derivatives & risk, ethics)
PM pathway share	≈ **30–35%** ≈ 7 scored sets

TRAP Topic weights here are from CFA Institute's 2025 pathway structure and prep providers. Check the official Level III page for any 2027 change.

Where the pathway marks come from ★★★★★

Reading	Share of pathway (est.)	Exam style	Priority
P6 Credit strategies	≈ 19%	Calculations (spread measures, CDS), strategy choice	★★★★★
P8 Institutional case study	≈ 15%	Essay: liquidity, asset allocation changes, ESG, risk	★★★★★
P4 LDI & index-based fixed income	≈ 14%	Calculations: immunisation, BPV hedge, contingent immunisation	★★★★★
P5 Yield curve strategies	≈ 13%	'Which trade?' + return estimates + currency	★★★★★
P2 Active equity: strategies	≈ 12%	Identify strategy / style / IC	★★★★
P7 Trade strategy & execution	≈ 12%	Benchmarks, implementation shortfall, algo choice	★★★★
P3 Active equity: portfolio construction	≈ 11%	Active share vs active risk, risk budget, long/short	★★★★
P1 Index-based equity	≈ 4%	Short explain questions	★★★

Fixed income (P4 + P5 + P6) ≈ **46%** of the pathway — prep providers call it the heaviest technical content

Essays favour 'determine the most appropriate strategy and justify'

Numbers to master: BPV hedge ratio, immunisation, rolldown return, spread duration, CDS price, implementation shortfall

How to read this summary ★★★☆☆

- Each subtopic starts with **In plain words** — read that first
- Formula boxes show the formula, then a worked example from the notes
- Red TRAP boxes are the mistakes that cost marks
- Page 1 lists every 5★ subtopic — the minimum to study

01 P1 Index-Based Equity Strategies

≈4% of PM marks · 8 subtopics · chapter priority ★★★☆☆

What this reading is about ★★★☆☆

In plain words: an index fund tries to earn exactly what an index (e.g. the S&P 500) earns — no more, no less. This reading covers: (1) factor-based indexes, (2) ways to get index exposure (funds, derivatives, own portfolio), (3) how to build the portfolio (full replication, sampling, optimisation), and (4) tracking error and its causes.

LOS	Topic
P1.a	Factor-based vs market-cap-weighted indexing
P1.b/c	Approaches to index-based investing (pooled, derivatives, separately managed)
P1.d	Full replication vs stratified sampling vs optimisation
P1.e	Causes and control of tracking error
P1.f	Sources of return and risk (attribution, securities lending, proxy voting)

Single-factor vs multifactor models (Fama-French five) ★★★★☆

In plain words: a 'factor' is a common driver of stock returns. CAPM says only one factor matters — the market (measured by beta). Multifactor models say returns depend on exposure to several factors.

Fama-French (2015) factor	How it is measured
1. Market risk premium	Beta (standardised covariance with market return)
2. Firm size	Market value of equity
3. Book-to-market	Shareholders' equity ÷ market value of equity
4. Operating profitability	Operating income ÷ beginning shareholders' equity
5. Investment intensity	Growth rate of total assets

Other common factors: growth, value, size, yield, momentum, quality, volatility

Factor indexes let investors **add to or replace** a cap-weighted index based on their view of future factor returns

Usually packaged as **ETFs**

Factor-based strategies: passive or active? ★★★★☆

Passive side: rules-based, transparent, replicable

Active side: choosing to emphasise some factors over the market portfolio is an active decision; managers may exploit out-of-favour factors

Costs: **lower** than traditional active funds, **higher** than large cap-weighted index funds (higher fees and commissions)

Transparency lets others copy the strategy → their buying moves prices → extra factor return can shrink or disappear

Managers often use several benchmarks (factor + cap-weighted) → can raise tracking error

TRAP A factor index fund is passive in **IMPLEMENTATION** but active in **FACTOR CHOICE**.

Three types of factor-based strategy ★★★★★

Type	Examples	Notes
Return-oriented	Dividend yield (high yield or dividend growth); momentum (overweight recent winners vs benchmark); fundamentally weighted (dividends, sales, income)	Focus on factors that explained past return differences
Risk-oriented	Volatility weighting (weight = 1 ÷ price volatility); minimum-variance (Markowitz, with sector/country limits)	Simple, reduce risk — but rely on **past** data
Diversification-oriented	Equal weighting; maximum diversification (maximise weighted-average stock volatility ÷ portfolio volatility)	Spread risk widely

Quiz: a **small-cap, high P/E** strategy → **return-oriented**

TRAP Minimum-variance is **RISK** -oriented, not diversification-oriented. Equal weight is **DIVERSIFICATION** -oriented.

Three ways to get index exposure ★★★★★

Approach	Advantages	Disadvantages
Pooled: open-end mutual fund	Low transaction costs, convenient structure	Must sell stocks to meet redemptions → taxable gains passed to all holders; traded only at close
Pooled: ETF	Trades intraday; redemptions met **in kind** by authorised participants → no forced sales, fewer taxable gains	Commissions + bid-ask spread; some ETFs illiquid in secondary market
Derivatives (futures, options, swaps)	Quick, cheap change of factor exposure; liquid markets; easy leverage	Expire → must roll; position limits; specialty needs may not exist; OTC counterparty risk; **basis risk** → tracking error
Separately managed portfolio	Full control, customised	Needs updated index data, trading & accounting systems, broker relationships for program trading, compliance systems

Overlay type	Use
Completion overlay	Return portfolio to index risk (e.g. match index beta)
Rebalancing overlay	Cheaply match index reconstitution (adds/deletes)
Currency overlay	Adjust FX risk of foreign holdings

Building the portfolio: replication, sampling, optimisation ★★★★★

Method	How	Pros	Cons / best for
Full replication	Hold every index stock at index weight	Closest match to index (before costs)	Costly with many / illiquid stocks; must reconstitute and rebalance. Best: **few, liquid, similar** stocks
Stratified sampling (cell matching)	Split index into strata (cells) by size, sector, country, dividend yield, momentum…; sample stocks in each cell; weight to match index exposures	Cheaper; keeps factor exposures	Strata must be **mutually exclusive and exhaustive**; more criteria → lower tracking error
Optimisation	Mean-variance model minimises tracking error subject to constraints (min stocks, style tilt, min cap)	Lower tracking error than sampling; uses **covariances** directly	Based on historical relationships that change; costly to maintain; result may not be mean-variance efficient
Blended	Replicate large liquid stocks; sample/optimise thin small caps	Best of both	E.g. Wilshire 5000

Optimisation can be run inside each stratum (combined with sampling)

Many heterogeneous, thinly traded small caps → sampling or optimisation

TRAP Tracking error vs sample size: **FIRST FALLS, THEN RISES** — the last stocks added are illiquid and costly (quiz answer C).

Tracking error: causes and control ★★★★★

In plain words: tracking error = standard deviation of (portfolio return − index return). For an index manager, low tracking error is the skill — the goal is to match, not beat.

```
Tracking error = SD(R_portfolio - R_index)
```

Cause	Why
Management fees	Index has no fees
Trade commissions	Index trades for free
Sampling	Not all stocks held
Intraday trading	Index uses closing prices
Cash drag	Cash held for flows; lowers return in rising markets, helps in falling markets

Control = trade-off: full replication's higher trading costs vs sampling's tracking error

Must still trade for inflows, outflows and dividend reinvestment

Index futures on idle cash reduce cash drag

Quiz: tracking error is a better skill measure than excess return because the goal is to **match** the index

Sources of return and risk ★★★★☆

Item	Point
Attribution analysis	Finds sources of tracking error so the manager can reduce them (details in core Performance Evaluation reading)
Securities lending	Lend stocks for fee income (or return on collateral) → offsets fees → **lowers** tracking error
Proxy voting	Index managers still have a fiduciary duty to vote; good governance can raise index returns; costly across many firms → many use **proxy-voting services**

Corporate governance and investor activism matter for index managers too

02 P2 Active Equity Investing: Strategies

≈12% of PM marks · 17 subtopics · chapter priority ★★★★☆

What this reading is about ★★★☆☆

In plain words: active managers try to **beat** a benchmark. This reading lists the ways they do it (fundamental vs quantitative; bottom-up vs top-down; factor, activist, stat arb, event-driven), how each strategy is built, its traps, and how to classify a manager's **style**. Learn the vocabulary, the steps, and the pros and cons.

LOS	Topic
P2.a	Fundamental vs quantitative
P2.b / c	Bottom-up and top-down strategies
P2.d	Factor-based active strategies
P2.e	Activist strategies
P2.f	Statistical arbitrage & market microstructure (plus event-driven)
P2.g	Building a fundamental strategy
P2.h	Building a quantitative strategy
P2.i	Style classification

Fundamental vs quantitative approaches ★★★★★

In plain words: a fundamental manager decides by **opinion** (judgment after research). A quant manager decides by **rules** found in historical data. Both may use the same data and tools — the difference is how the final buy/sell decision is made.

	Fundamental	Quantitative
Style	Subjective, discretionary	Objective, systematic (rules)
Decision based on	Analyst's research and judgment	Model of past data
Focus	Company: financials, business model, management, industry position → intrinsic value	Factors that predict returns across many stocks (P/E, size, D/E, sector, momentum)
Forecast	Intrinsic value and return of each firm	Expected return from factor exposures
Portfolio	Fewer, larger positions; size by conviction within mandate limits	Many smaller positions; optimiser maximises alpha or information ratio
Main risk	**Company** level: value wrongly estimated, or market never corrects the mispricing	**Portfolio** level: factors fail to perform as the model predicts
Rebalancing	Continuous, whenever opinion changes	Automatic, at set intervals (monthly, quarterly)

Fundamental managers may use screens, DCF and regression; quants may use financial-statement data

Quiz: screening for low P/B can be used by **both** types (answer C)

Bottom-up vs top-down ★★★★★

	Bottom-up	Top-down
Information used	Single companies (profitability, momentum, brand, moat, management, governance)	Macro, policy, demographics — variables that affect many companies
Selects	Best individual stocks	Best countries, sectors, styles, themes
Tools	Stock purchases	Broad ETFs, index derivatives

Blends: top-down strategist sets country/sector weights → bottom-up managers pick stocks inside them

Or: bottom-up picks stocks, then a top-down **derivatives overlay** removes unintended macro exposures

Quant bottom-up: measurable links between firm data (P/E) and return. Fundamental bottom-up: also qualitative traits (brand, competitive advantage, management, governance)

Bottom-up value sub-styles ★★★★★

In plain words: value investors buy stocks priced **below** their estimated intrinsic value.

Sub-style	Looks for
Relative value	Low P/E, P/B vs peers/industry with no clear reason
Contrarian	Going against sentiment, e.g. depressed cyclical stocks with low or negative earnings
High-quality value	Value **plus** financial strength, good management, proven profitability ('Warren Buffett' style)
Income investing	High dividend yield and dividend growth
Deep value	Extremely low value vs assets (low P/B), often due to distress
Restructuring & distressed debt	Buy before or during expected bankruptcy; value released via restructuring or liquidation
Special situations	Mispricing from corporate events: divestitures, spin-offs, mergers

TRAP Undervalued vs **BANKRUPTCY LIQUIDATION VALUE** = restructuring/distressed debt (quiz B), not deep value.

Bottom-up growth sub-styles & worked selection example ★★★★★

In plain words: growth investors buy firms whose sales, earnings or cash flows will grow faster than the industry or market. They care less about high multiples and more about the source and persistence of growth.

Consistent long-term growth | Short-term earnings momentum

GARP (growth at a reasonable price) — uses the PEG ratio

```
PEG = P/E ÷ expected EPS growth (in % terms, e.g. 15 ÷ 10 = 1.5)
Example (match each stock to one strategy, each used once):
TW — lowest P/B 0.75 (only stock below 1) → deep value
NB — lowest PEG 15/10 = 1.5 → GARP
SO — P/E 20 vs sector 30 = 0.67 → relative value (TW's 1.5/8 is lower, but TW is needed for deep value)
TO — dividend yield 1.08/12 = 9% → income investing
```

Top-down strategies & smart beta ★★★★☆

Dimension	How
Country / geography	Overweight attractive markets via ETFs, futures
Industry sector	Overweight / underweight sectors
Volatility	VIX futures, variance swaps, straddles (e.g. GARCH forecast: buy options when implied vol low, sell when high)
Thematic	New technologies, regulation, economic cycles; long-term (cloud, blockchain, clean energy) or short-term (currency effect of a vote)
Equity style rotation	Rotate into styles (factors) expected to outperform

Country/sector views can be checked bottom-up by adding up company values

Smart beta: rotate exposures into factors (betas) expected to earn more — a top-down idea; made easy by focused ETFs and structured products

Quiz: GARP on S&P 500 stocks = **bottom-up**, not top-down

Factor-based strategies: rewarded factors & hedged portfolio ★★★★★

In plain words: a factor is a stock trait linked to returns (e.g. small size, high book-to-market). **Rewarded** factors earn a long-term risk premium; **unrewarded** ones do not. A factor must make intuitive sense — otherwise heavy backtesting finds false patterns.

```
Hedged portfolio approach (Fama-French):
1. Rank universe by the factor (size → market cap)
2. Split into quantiles (deciles 10%, quintiles 20%,
quartiles 25%)
3. Long best quantile, short worst (size: long smallest
10%, short largest 10%)
4. Track the long/short return = the factor's return
```

Drawback	Meaning
Middle quantiles ignored	Best stocks may sit in the middle
Assumes linear factor–return link	Non-linear effects missed
False diversification	Several highly correlated factors
Needs short selling	Not always allowed
Not a pure factor	Has exposures to other factors

Factor-mimicking portfolio: theoretical, dollar-neutral, exposure of 1 to the chosen factor and 0 to others; hard to build (liquidity, shorting)

Long-only investors **tilt** toward factors; modest tilts = low tracking error = enhanced indexing

Real value added = finding factors that predict the future

TRAP Value factor: go **LONG HIGH** book-to-market (cheap), short low B/M. Using quartiles instead of deciles is **NOT** an error (quiz C).

Factor timing (style rotation) worked example ★★★★☆

In plain words: some factors work better in some conditions. Regress factor returns on a suspected driver to find when to hold each factor.

```
Driver: ISurprise = actual central bank rate - rate
priced in Eurodollar futures the day before (high =
hawkish surprise)
Current month: beta (market) coefficient -0.5 → high-beta
stocks lose on upward surprise; size and value no signal
Next month: size -1.0 (small caps underperform after
hawkish surprise); value also negative; beta ≈ +0.7
Issue: beta sign flips (- now, + next month) → need to
know how long the effect lasts before trading on it
```

Activist strategies ★★★★★

In plain words: activists buy a minority stake and push management for changes (strategy, capital structure, ESG, governance) to raise the share price. Example: Carl Icahn.

```
Process: screen → buy stake (usually < 10% of votes) →
public proposal (open letter) → threaten proxy contest →
launch proxy contest → keep negotiating
```

Activist tactics	Management defences
Seek board seats (then change management)	Multi-class shares (founders hold extra votes)
Open letters, meetings, rally other shareholders	**Poison pill**: existing holders buy shares at a discount → dilutes activist
Proposals at the AGM	**Staggered board**: only part re-elected each year
Financial restructuring: more dividends, buybacks	
Cut excessive pay; sue for breach of fiduciary duty	
Media campaign; break up an inefficient conglomerate	

Roots: 1970s–80s corporate raiders; activist AUM more than doubled 2007–2015; campaigns rose four-fold

Targets: slower growth than market, negative price momentum, weak governance

Impact: better growth, profitability, governance, but **higher debt**; Sharpe ratios only slightly above market; prices rise before, most on the announcement day, modestly over the next month

TRAP Activists do **NOT** buy a majority stake (quiz A).

Statistical arbitrage & market microstructure ★★★★★

In plain words: stat arb uses price and volume data to profit when related prices drift apart and then come back (mean reversion).

```
Pairs trading: two historically correlated stocks in one
industry
Spread = ln(P_A / P_B)  (log gives a more stable signal)
Spread > moving average + 2 SD → SELL the spread (short
A, long B)
Spread < moving average - 2 SD → BUY the spread
Profit if the gap closes; risk = relationship broken for
good
```

Market microstructure arbitrage: supply/demand imbalances lasting milliseconds; needs limit order book analysis and **high-frequency trading** (quiz B)

Stat arb and event-driven are usually quantitative but may include judgment

Event-driven strategies & risk arbitrage ★★★★★

In plain words: event-driven managers profit from mispricing around corporate events: M&A, earnings, restructuring, buybacks, special dividends, spin-offs. Unlike activists, they do **not** cause the event.

Merger type	Risk arb trade	Profit / risk
Cash deal	Buy target after announcement (trades just below deal price)	Earn the gap when deal closes; big loss if deal fails
Share-for-share	Buy target, short acquirer in the exchange ratio	Same

Key skill: estimate probability of failure, deal length and annualised premium

```
Case: Formby (cash-rich tech, slowing growth, no dividend) → activist: push for dividends/buybacks
Parmeon (1-for-1 share deal, regulatory risk) → event-driven: buy target, short Parmeon 1:1
Baron (distressed commodity trader) → distressed debt: buy bonds if market too pessimistic
NOT a pairs trade for Baron — correlation broke for fundamental reasons, no mean reversion
```

Creating a fundamental strategy & its pitfalls ★★★★★

```
1. Define universe per mandate; state the market opportunity (thesis)
2. Prescreen to a manageable list (value manager drops high P/E)
3. Analyse industry, competitive position, financial reports
4. Forecast (cash flows or earnings)
5. Convert forecasts to valuations
6. Build portfolio with desired risk; add top-down views/constraints
7. Rebalance; sell disciplines: target prices (take profit) and stop-losses
```

Pitfall	Meaning
Behavioural biases	Confirmation, illusion of control, availability, loss aversion, overconfidence, regret aversion
Value trap	Low multiple **justified** by deteriorating business → price keeps falling. Need a trigger for revaluation
Growth trap	Growth already in (or over) the price: price assumes 15%, firm delivers 12% → P/E and price fall

Sell disciplines and stop-losses fight the bias to hold losers

Creating a quantitative strategy: steps & data ★★★★★

```
1. Define market opportunity (thesis)
2. Acquire and process data (most time-consuming)
3. Backtest
4. Evaluate (out-of-sample)
5. Construct portfolio
```

Data type	Content
Company mapping	Track firms over time and across vendors (mergers, bankruptcies, identifiers)
Company fundamentals	Demographic, price, financial data (e.g. Bloomberg)
Survey data	Earnings forecasts, macro, sentiment, fund flows
Unconventional data	Satellite images, news sentiment, supply chains, events — machine learning helps

Backtesting: factor scores and information coefficient (IC) ★★★★★

In plain words: the IC is the correlation between today's factor scores and **next period's** returns across stocks. Positive and significant = the factor predicts.

```
Standardised factor score (z) = (stock value - mean) / SD
E/P 7%, mean 3%, SD 5% → (7 - 3)/5 = 0.8
Pearson IC = correlation(factor score, subsequent return) — monthly 5–6% is very strong; hurt by outliers
Spearman Rank IC = correlation of the ranks → robust to outliers
```

```
9-stock example: Pearson IC -0.99% (distorted by stock A: negative score, +10.06% return)
Spearman Rank IC +40% → high E/P does predict
Long top 3 (G, H, I): (3.02 + 3.05 + 5.20)/3 = 3.76%
Short bottom 3 (A, B, C): (10.06 - 0.6 - 0.5)/3 = 2.99%
Long/short = 0.77%; sample too small → test more data and periods
```

Multifactor model: weight factors by judgment or mean-variance optimisation; a factor adds little if correlated with others

Evaluate: **out-of-sample** tests; returns plus VaR and maximum drawdown

Portfolio construction: commercial **risk models** (variances, correlations) and **trading costs** (explicit + market impact) — same return → choose lower cost

TRAP Predictive power uses **SUBSEQUENT** returns, not contemporaneous. Outliers → **SPEARMAN** (quiz B).

Pitfalls in quantitative investing ★★★★★

Pitfall	Meaning
Survivorship bias	Test only on firms still alive → results look too good
Look-ahead bias	Using data before it was available (December accounts to trade in January)
Data mining / overfitting	Search until something works — not real proof
Turnover	Limits stop the strategy being followed
Stock not available to borrow	Shorts cannot be placed
Transaction costs	Erode backtested returns
Quant crowding	Many quants hold the same trades → mass exit, margin calls, fire sales (August 2007 quant meltdown in a calm market). Warning sign: high correlation of short interest and price momentum

Style classification: holdings-based (Morningstar style box) ★★★★★

In plain words: style analysis says what kind of manager someone really is (large value, small growth…). Good style groups have high correlation inside and low correlation between groups.

Size	Value	Blend (core)	Growth
Large (top 70% of market cap)			
Mid (next 20%)			
Small (rest)			

Value score (weight)	Growth score (weight)
Forward P/E 50%	Long-term projected EPS growth 50%
P/B 12.5%	Historical EPS growth 12.5%
P/S 12.5%	Sales growth 12.5%
P/CF 12.5%	Cash flow growth 12.5%
Dividend yield 12.5%	Book value growth 12.5%

Each stock gets value and growth scores 0–100; net style = growth − value: strongly negative → value; strongly positive → growth; near 0 → core

Morningstar: a **stock** with no bias = **core**; a **fund** with no bias = **blend**

Columns hold about equal numbers of stocks; a stock can be part value, part growth (e.g. 60% to value index, 40% to growth)

Returns-based style analysis, self-identification, pros & cons ★★★★★

```
Regress fund returns on passive style indexes, slopes sum
to 1 (and ≥ 0)
R_fund = b1 LargeGrowth + b2 SmallGrowth + b3 SmallValue
+ b4 LargeValue
b1 = b2 = 0, b3 = 0.15, b4 = 0.85 → value manager, mostly
large-cap value
```

	Holdings-based	Returns-based
Shows	**Current** exposure (if holdings up to date)	**Average** exposure over the period
Data	Detailed holdings; costly; needs history for averages	Only returns; cheap, quick
Weakness	Misses derivatives overlays unless disclosed; holdings may be stale/window-dressed	Lags style changes; wrong index choice; collinear indexes; black box
Strength	Precise, explains each stock	Captures effect of derivatives and all positions

Self-identification: manager's stated style — compare with both analyses; for long/short, market-neutral, short bias the prospectus is the key source

TRAP Undisclosed derivatives overlays mislead **HOLDINGS-BASED** analysis only (quiz A).

03 P3 Active Equity Investing: Portfolio Construction

≈11% of PM marks · 19 subtopics · chapter priority ★★★★☆

What this reading is about ★★★☆☆

In plain words: P2 was about **finding** ideas; P3 is about **turning ideas into a portfolio**: how big each bet is, how far to move from the benchmark (Active Share, active risk), how to spend a risk budget, how size and trading costs limit a strategy, and when to use shorts.

LOS	Topic
P3.a	Investment philosophy: sources of active return, building blocks, fundamental law
P3.b	Approaches; objectives and constraints
P3.c	Active Share vs active risk
P3.d	Risk budgeting (absolute and relative)
P3.e	Heuristic vs formal risk constraints
P3.f	AUM, position size, liquidity, turnover
P3.g	The well-constructed portfolio
P3.h	Long-only, long extension, long/short, market-neutral

Active return and its three sources ★★★★★

In plain words: active return = portfolio return − benchmark return. It is positive if you overweight winners and underweight losers. Active portfolios aim to beat the benchmark **after all costs**.

$$R_A = \Sigma (w_{pi} - w_{\beta i}) \times R_i$$

$$\text{Ex post: } R_A = \underbrace{\Sigma (\beta_{pk} - \beta_{\beta k}) \times F_k}_{\text{(factor exposure)}} + \underbrace{\alpha}_{\text{(skill)}} + \varepsilon$$

(idiosyncratic = luck)

Source	What it is
1. Rewarded factor exposure	Long-term exposures that differ from the benchmark to factors with positive risk premiums: market (beta), size, value, liquidity, momentum
2. Alpha	Tactical bets on mispriced securities, sectors, factors — return not explained by factor exposure = **skill**
3. Idiosyncratic risk	Return from concentrated positions not due to factors or skill = **luck**

The building blocks of portfolio construction ★★★★★

Block	Meaning	Main link
1. Factor weightings	Over/underweight rewarded factors vs benchmark. Now cheap via rules-based index products → static factor exposure is **no longer alpha**	Source 1
2. Alpha skills	**Factor timing** of rewarded factors (value vs growth) or unrewarded ones (countries, sectors, commodities, single stocks — 'thematic' for discretionary managers)	Source 2
3. Position sizing	Balance conviction against idiosyncratic risk. Affects all three sources, most of all idiosyncratic. Many small positions → lower volatility	Source 3
4. Breadth of expertise	Integrates the three; broader expertise → more consistent active return	All

Factor-oriented manager: many assets → little idiosyncratic risk. Stock picker: concentrated → takes idiosyncratic risk on purpose

Fundamental law of active management ★★★★★

In plain words: expected outperformance grows with skill (IC), with the number of independent bets (breadth), with risk taken, and with how freely insights reach the portfolio (TC).

```
E(R_A) = IC × √BR × σ_A × TC
IC = information coefficient (skill) ; BR = breadth
(independent decisions per year)
σ_A = active risk ; TC = transfer coefficient (0–1)
Quiz: 10 decisions per month → BR = 120 ; IC 0.1, σ_A 5%,
TC 0.5
E(R_A) = 0.1 × √120 × 5% × 0.5 = 2.74% (answer C)
```

Single factor, single metric → decisions not independent → low breadth

Several factors, several metrics each → more independent decisions → higher breadth

Fewer constraints (e.g. allowing shorts) → higher TC → higher information ratio

TRAP Breadth must be **ANNUAL** — multiply monthly decisions by 12.

Approaches to portfolio construction ★★★★☆

Approach	Factor weightings	Alpha skills (timing)	Position sizing	Optimisation
Systematic bottom-up	Diversified rewarded factors	Rule-based signals	Many small positions	Formal optimiser common
Systematic top-down	Factor/macro rules	Some rule-based timing	Diversified	Formal
Discretionary bottom-up	Implicit via stock choice	Security selection	Concentrated	Less formal
Discretionary top-down	Macro/thematic bets	**Most factor timing**	Concentrated in themes	Less formal

Systematic = rules; discretionary = judgment. Bottom-up = stock data; top-down = macro data

Discretionary managers: more factor timing, more concentrated, less formal optimisation

Quiz: factor timing most likely by **discretionary top-down**

Objectives and constraints ★★★★☆

In plain words: building a portfolio is an optimisation: one goal (objective) plus limits (constraints), stated in absolute or benchmark-relative terms.

	Absolute	Relative
Objective	Maximise Sharpe ratio	Maximise information ratio
Constraint	Maximum volatility	Maximum active risk (tracking error)

Other objectives: minimise volatility, downside risk or drawdowns

Maximise exposure to rewarded factors (size, value, momentum)

Maximise a custom trait set by a discretionary manager (e.g. a deep-value metric)

Heuristic weighting: rank stocks by a trait (e.g. low P/B) — less scientific

Active Share ★★★★★

In plain words: Active Share measures how different your **holdings** are from the benchmark's. 0 = identical; 1 = no stock in common. Benchmark-aware managers keep it low; benchmark-agnostic managers let it be high.

```
Active Share = ½ × Σ |w_Pi - w_βi|  (= total overweights =
total underweights)
Overlap with benchmark = 1 - Active Share
Benchmark: A, B, C, D 25% each (universe A–F)
Portfolio = benchmark → 0 ; portfolio only in E, F → ½ ×
(4 × 25% + 100%) = 1
Portfolio B 10%, C 25%, D 25%, E 40% → ½ × (25 + 15 + 0 +
0 + 40)% = 0.40 → overlap 0.1 + 0.25 + 0.25 = 0.60
```

Only benchmark stocks, same benchmark → **fewer** stocks = more concentrated = **higher** Active Share

Fee check: Active Share 0.2 at same fee as 0.5 → expensive (closet indexing)

Manager **fully controls** Active Share (sets weights)

Active risk & how it differs from Active Share ★★★★★

In plain words: active risk (tracking error) = how much active return jumps around from period to period. It depends on **correlations**, so the manager cannot fully control it.

```
Active risk σ_A = SD(R_P - R_β)
σ²_A = active factor risk (variance from β_P - β_β) + active
idiosyncratic risk (incl. alpha)
```

	Active Share	Active risk
Measures	Holdings difference	Volatility of active return
Uses correlations?	No	Yes
Realised vs predicted	—	Realised (history) and predicted (needs forecasts)
Control	Full	Partial
Swap pharma A → similar pharma B	Up	Hardly up (quiz A)
Swap pharma → consumer discretionary	Up	Up

High net factor exposure → high active risk

No net factor exposure → active risk comes only from Active Share

Active risk from Active Share falls as the number of stocks rises (inversely proportional)

Active risk rises with factor and idiosyncratic risk

TRAP Factor bets **ALWAYS** raise Active Share; high Active Share does **NOT** always mean factor bets (stock A for stock B with the same exposures).

Classifying managers with Active Share and active risk ★★★★★

Manager	Active Share	Active risk	Typical limits
Pure indexer	~0	~0	None
Closet indexer	Low	Low	Small deviations allowed
Diversified multi-factor	Moderate	Moderate	Moderate sector deviation; low single-stock risk (many positions)
Sector rotator	Moderate	**High**	Large sector deviations allowed
Concentrated stock picker	**High**	High	Large single-stock deviations; may have sector weights fixed

Worked example: A = sector rotator (large sector deviations) · D = concentrated stock picker (large security risk, no sector deviation) · C = closet indexer (low target active risk) · B = diversified multi-factor (moderate everything, low single-stock risk)

Tip: start with the most obvious manager, then eliminate

Risk budgeting: the four steps ★★★★☆

In plain words: a risk budget shares out the total risk you accept among positions or factors, like a spending budget shares out money.

```
1. Choose the measure: absolute (total-return mandate,
e.g. long/short vs cash + margin) or relative (beat an
index)
2. Understand what drives risk (rewarded factors?
sectors? single stocks?)
3. Set the overall risk level
4. Allocate risk to positions/factors efficiently
```

Absolute risk: contribution of an asset and of a factor ★★★★★

```
Contribution of asset i to variance: CVᵢ = Σⱼ wᵢ wⱼ
Cov(i,j) = wᵢ × Cov(rᵢ, rₚ)
σₚ² = Σ CVᵢ
Worked: wA 0.20, wB 0.30, wC 0.50 ; Var(A) 0.05, Cov(A,B)
0.0067, Cov(A,C) 0.0013
CV_A = 0.2×0.2×0.05 + 0.2×0.3×0.0067 + 0.2×0.5×0.0013 =
0.002532
σₚ² = 0.002532 + 0.001998 + 0.002880 = 0.00741 → A's
share = 34.2%
```

```
Contribution of factor i: CVᶠᵢ = Σⱼ βᵢ βⱼ Cov(Fᵢ, Fⱼ) ;
σₚ² = Σ CVᶠ + unexplained
Market β 0.892, size β -0.283 ; Var(M) 0.00178, Cov(M,S)
0.00042 ; σₚ = 3.74%
Market: 0.892×0.892×0.00178 + 0.892×(-0.283)×0.00042 =
0.00131
σₚ² = 0.0374² = 0.0013988 → market explains 93.7%
Size -0.00007 → explained 0.00124 → unexplained 0.00016 =
11.4%
```

Country or sector contribution = its weight × covariance with the **portfolio** (quiz C — not with the global market, not correlation)

Shares can add to over 100% when a factor contributes negatively

Multi-factor manager: most risk explained by rewarded factors, little unexplained

Relative (active) risk contribution ★★★★☆

```
CAVᵢ = (wₚᵢ - wᵦᵢ) × Σⱼ (wₚⱼ - wᵦⱼ) × Cov(RAᵢ, RAⱼ)
Σ CAVᵢ = active variance AVₚ
```

Same idea as absolute, but uses **active weights** and covariances of **active returns**

Example: benchmark 50% index A + 50% index B; portfolio holds A, B and cash. CAV_A 0.000233, CAV_B −0.000233, CAV_cash 0.001296 → cash = **100%** of active variance (calculation not tested)

Cash has low SD but **high active risk** (twice the indexes') because it has low correlation with the benchmark

Active returns of A and B correlate −1: when one beats the 50/50 benchmark the other must lag

Can be done by country, sector or factor; split into active factor variance + unexplained

Manager	Main risk sources
Factor diversified	Balanced rewarded factor exposures; low idiosyncratic and sector risk
Sector rotator	Concentrated **sector** and **cash** bets dominate

Choosing the right level of risk ★★★★☆

Issue	Why it limits risk
Implementation constraints	Short/leverage limits stop full over/underweights; liquidity costs rise with active risk → **information ratio falls**
Limited diversification	Concave efficient frontier: high-risk targets run out of good ideas → **Sharpe ratio falls**
Leverage	Moves you up the straight capital allocation line for one period, but too much lowers compounded return

```
R_g ≈ R_a - σ²/2   (σ is squared, so leverage eventually
lowers geometric return)
-2% then +2%: 0.98 × 1.02 = 0.9996 ; -20% then +20%: 0.8
× 1.2 = 0.96
```

Heuristic vs formal risk constraints ★★★★★

Type	Examples	Needs return-distribution forecast?
Heuristic (experience, good practice)	Limits on single positions, sectors, regions; leverage; illiquidity; turnover	No
Formal (statistical)	Volatility, active risk, skewness, drawdown, VaR family	**Yes** → estimation error (quiz B)

VaR measure	Meaning
CVaR (expected shortfall / expected tail loss)	Expected loss **given** VaR is exceeded
Incremental VaR (IVaR)	Change in VaR from **adding a new** position
Marginal VaR (MVaR)	Change in VaR from a **very small** change in a position

Leverage magnifies wrong risk estimates; tail events (early 2000s, 2008) force sales at bad prices

Volatility spikes → managers tighten risk controls

Benchmark-agnostic absolute-return manager → watches **drawdown**, not active risk; market-neutral manager → keeps **absolute volatility** in a range

Few positions → higher estimation error → formal measures harder to use

AUM, liquidity, turnover and market impact ★★★★★

In plain words: big trades move prices against you (market impact). This hidden cost eats alpha and grows with fund size, small caps, turnover and 'informative' trades.

Impact ↑ with: higher AUM vs stock market cap · smaller caps · higher turnover · shorter horizon · trades that carry information

Slippage = execution price − bid/ask midpoint when the order was entered

Slippage facts: 1. usually > explicit costs · 2. higher for small caps · 3. **not necessarily** higher in emerging markets · 4. much higher in volatile markets

Small-cap manager as AUM grows: close to new money, tell investors the strategy may change, limit AUM, diversify, cut turnover, change trading

Beware new funds raised on a track record that cannot scale; large-cap funds can hold more AUM

```
Small cap $5bn, 1% traded daily → ADV $50m; limit 10% of
ADV → max $5m
$1bn fund → max weight 0.5% (maybe too small for a
concentrated strategy)
```

```
Scale: $200m fund, 350 stocks, index $20tn, 1,000 stocks,
small ones trade 1.5% a day
Rule 1 no stock with index weight < 0.02% → smallest
stock = 0.0002 × 20tn = $4bn
ADV = 0.015 × 4bn = $60m ; Rule 3 max 5% of ADV → $3m
Rule 2 max weight = lesser of 10 × 0.02% = 0.2% and 0.02%
+ 1% = 1.02% → 0.2%
Impaired when AUM > 3m / 0.002 = $1.5bn
```

TRAP Higher market impact comes with **HIGHER** AUM and turnover, and **SMALLER** caps (quiz C).

The well-constructed portfolio ★★★★☆

Clear philosophy + consistent process

Risk and structure as promised to investors

Desired exposures achieved efficiently

Reasonably low operating costs

Same exposures with **fewer** positions → more risk-efficient (more risk-management focus; no return guarantee)

Similar factor exposures → prefer lower absolute volatility and lower active risk

Similar risks, costs and skill → prefer **higher Active Share** (leverages alpha skill) — quiz B

Multi-manager equity sleeve: combine managers so a missing factor exposure is filled by a specialist

Long-only vs long/short: factors in the choice ★★★★★

In plain words: short selling = borrow a stock, sell it, buy back later cheaper. It lets a manager act on **negative** views. Any portfolio that can short is 'long/short'.

Factor	Favours long-only because...
Long-term risk premiums	Premiums are earned by being net **long**; long-term shorts lose (short horizons may want shorts)
Capacity and scalability	Long capacity set by liquidity; short capacity set by **stock available to borrow** → lower
Limited legal liability	Long loss capped at price paid; short loss **unlimited**
Regulations	Some countries ban short selling
Transactional complexity	Shorts need borrowing, collateral, recall risk, **prime broker** (extra counterparty risk)
Costs	Higher fees and operating costs for long/short
Personal ideology	Moral objection; doubt managers can short well; dislike leverage

Long/short structures: gross, net, 130/30, market-neutral ★★★★★

```
Gross exposure = long + |short| (% of capital) ; Net =
long - short
$100m capital, $80m long, $30m short → gross 110%, net
50%
Cash $50m = $20m uninvested + $30m short-sale proceeds
130/30 long extension: long 130%, short 30% → net 100%,
gross 160%
Market-neutral: long $200m × β 0.9 = 180 ; short $150m ×
β 1.2 = 180 → net β = 0
Equal long and short size → gross = 2 × long, net = 0
```

Long extension: net fixed at 100%; manager has no discretion over gross/net; for investors wanting full market exposure plus some shorting

Market-neutral: removes market risk where the manager has no edge; low volatility, low correlation → used for **diversification**; zero beta hard to hold as correlations change

Built via **pairs trading**; quantitative pairs trading = stat arb

Long/short: benefits and drawbacks ★★★★★

Benefits	Drawbacks
Express negative views fully (long-only max underweight = benchmark weight) → higher TC → higher IR	Losses unlimited (prices have no ceiling); **reduces exposure to market risk premium**
Use short proceeds to **gear into high-conviction longs** (quiz A)	Leverage magnifies losses as well as gains
Remove market risk → diversifier	Borrow costs can become very high (hard-to-borrow stocks)
Better control of factor exposures (factors are long/short by nature)	Collateral calls force sales; **short squeeze**; lenders recall shares

TRAP High leverage is **BOTH** a benefit and a drawback — not the best answer for 'benefit'.

04 P4 Liability-Driven & Index-Based Fixed Income

≈14% of PM marks · 18 subtopics · chapter priority ★★★★★

What this reading is about ★★★☆☆

In plain words: part 1 (LDI) — hold bonds so you can pay known future bills (liabilities) whatever rates do: immunisation, duration matching, closing a 'duration gap' with futures, swaps and swaptions, and the risks left over. Part 2 — copying a bond index cheaply, other ways to get passive bond exposure, and choosing a bond benchmark.

LOS	Topic
P4.a	Managing a single liability (immunisation)
P4.b	Single vs multiple liabilities; ways to implement
P4.c	Liability strategies under rate scenarios (futures, swaps, swaptions, collars)
P4.d	Risks of managing against liabilities
P4.e	Bond indexes and the challenges of indexing
P4.f	Passive bond exposure: funds, ETFs, total return swaps
P4.g	Selecting a bond benchmark

Duration refresher: Macaulay, modified, price vs reinvestment risk ★★★★★

In plain words: immunisation = manage a bond portfolio so its value at a set date is predictable, so a known bill can be paid. The perfect tool is one default-free zero-coupon bond maturing on the bill date — no coupons to reinvest, nothing to sell early.

Term	Meaning
Macaulay duration	PV-weighted average time to receive cash flows. Zero-coupon = its maturity; coupon bond < maturity
Price risk	Uncertain proceeds if you must **sell** before maturity
Reinvestment risk	Uncertain value of coupons **reinvested** to the horizon
Balance point	At horizon = Macaulay duration the two risks cancel — for one small, immediate, parallel rate change
Modified duration	Macaulay ÷ (1 + periodic yield): slightly better for **immediate price change**

Rates up: price loss offset by higher reinvestment income · Rates down: price gain offset by lower reinvestment income

Macaulay is slightly better for immunisation **timing**

$$\%\Delta Value \approx (-MD \times \Delta Y) + (\tfrac{1}{2} \times C \times \Delta Y^2)$$

Portfolio statistics from aggregate cash flows ★★★★☆

In plain words: for liability work, compute yield, duration and convexity from the portfolio's **combined cash flows**, not as market-value-weighted averages of each bond. (You will not compute these on the exam — understand how to use them.)

1. List every portfolio cash flow by six-month period
2. Add cash flows in each period
3. Portfolio IRR = rate that equates them to current market value
4. PV each cash flow at the IRR (sum = market value)
5. Weight w = PV ÷ total PV
6. Macaulay duration = $\Sigma\ t \times w$ (÷ 2 for years)
7. Dispersion = weighted variance of timing around duration
8. Convexity = $\Sigma\ [t\ (t + 1)\ w] \div (1 + \text{periodic IRR})^2$

Flat curve: same answer either way

Upward-sloping curve: portfolio IRR and duration are **higher** than the averages (long bonds count for longer)

Goal = earn the initial **portfolio IRR**, not average YTM

Averages are used in practice as approximations — they add error

Dispersion and convexity show exposure to structural risk from curve shifts and twists

Effects of yield curve changes on a barbell immunising a bullet ★★★★★

In plain words: a 5-year liability, matched by a barbell of a short (S) and a long (L) bond with the same duration (M = 5). Assets have more convexity than the liability. What happens when the curve moves?

Curve move	PV assets vs PV liability	Result
Parallel up (large)	Assets fall less than PVL (convexity); IRR and discount rate rise equally	FVA > FVL → **success**
Parallel down (large)	Assets rise more than PVL	FVA > FVL → **success**
Steepening twist (L yield up, S down, M fixed)	PVA falls below PVL (long bond loses more)	Risk — may still work if portfolio IRR rises enough
Flattening twist (L down, S up)	PVA rises above PVL	Risk — portfolio IRR may fall too far
Positive butterfly (S and L up, M down)	PVA ↓, PVL ↑	**Significant** structural risk
Negative butterfly (S and L down, M up)	PVA ↑, PVL ↓	Favourable, not guaranteed (IRR may drop)

Structural risk: assets' durations spread out (S, L) while liability sits at one point (M)

Immunisation = **zero replication**: success if portfolio IRR tracks the yield of the replicating zero

Parallel shifts are **sufficient but not necessary** for success

Not buy-and-hold: coupon bond duration falls slower than the liability's (which falls one year per year) → rebalance

Rules for immunising a single liability; money duration ★★★★★

```
1. PV assets ≥ PV liability (exceptions if portfolio IRR
≠ liability discount rate)
2. Portfolio Macaulay duration = liability due date
3. Minimise convexity (least dispersion of asset cash
flows around the liability)
4. Rebalance regularly (trade-off: transaction costs vs
duration drift)
```

```
Money duration (BPV, PVBP, 'value of an 01') = MD × Value
× 0.0001
Matching BPVs combines rules 1 and 2
```

Lower dispersion → lower convexity → less structural risk; dispersion 0 = a zero-coupon bond = perfect match

TRAP +C is good for price, but immunisation is about reaching the **FUTURE VALUE** → minimise convexity (quiz: match Macaulay duration to due date = A).

Multiple liabilities: cash-flow matching, duration matching, contingent immunisation ★★★★★

Method	How	Notes
Cash-flow matching (dedication)	Zero or coupon bonds whose flows match each payment	Simple; govt bond portfolio allows legal **defeasance** (assets and debt off balance sheet); expensive
Duration matching	Match **BPV** of assets and liabilities	Flexible, cheaper; structural risk; BPV handles different sizes and discount rates
Contingent immunisation	Active management while surplus > 0; immunise before surplus hits 0	Needs significant surplus; hybrid active/passive

```
Multiple-liability rules:
1. PVA ≥ PVL  2. BPV_A = BPV_L
3. Asset dispersion & convexity greater than liabilities'
(but not by much)
4. Rebalance to keep BPVs matched
```

```
UK company retiring bonds: tender too costly (bonds held
by buy-and-hold investors); defeasance with gilts costs
GBP475m (too high)
→ duration-matching portfolio of high-quality corporates
(cheaper, no defeasance, may help rating)
Choose Portfolio B: BPV closely matches + convexity
slightly above liabilities
Expect very large parallel shifts → Portfolio C (much
higher convexity, weaker BPV match)
```

TRAP Single liability: minimise convexity. Multiple: convexity must **EXCEED** liabilities', then be minimised.

Closing a duration gap with futures (derivatives overlay) ★★★★★

In plain words: instead of selling and buying bonds, add bond futures to raise or lower asset BPV. Buying futures adds duration.

Treasury futures on 2-, 5-, 10-yr notes and 30-yr bond; seller delivers 100,000 par of a deliverable bond — the **cheapest-to-deliver (CTD)**; buyer pays futures price × conversion factor

CTD's duration sets the contract's BPV; **Ultra 10-year** narrows deliverables to durations near 10 years

```
Futures BPV ≈ BPV_CTD ÷ CF_CTD
N_f = (BPV_L - BPV_A) ÷ BPV_futures   (+ = buy, - = sell)
UK revised: gap opened from 13 to 218,517 - 203,456 =
15,061 after a positive butterfly
5-yr: 48.1650 / 0.9237 = 52.1436 → buy 289
10-yr: 86.7001 / 0.9169 = 94.5579 → buy 159
Liability duration 8.99 → 10-yr contract = closer match →
less structural risk
```

PVA = PVL need not hold after initiation; success if value tracks the replicating zero

Overfunding is common; big surplus → contingent immunisation

Contingent immunisation in practice ★★★★☆

Way to use the surplus	Comment
Invest whole portfolio in stocks	Most active; must liquidate fast if surplus shrinks
Only surplus in stocks / long options; rest immunised	Safer — only the surplus at risk
Active bond management: over/underhedge	Expect rates ↑ → underhedge (<100%); expect rates ↓ → overhedge (>100%)

```
Each 1/32 price move on 100,000 par = (1/32) ×
(100,000/100) = $31.25 per contract
```

Liquidity risk: must convert to immunised portfolio quickly; short options and futures can lose more than the surplus

Success → surplus grows, cost below immunising; failure → must immunise at a higher cost

TRAP Contingent immunisation = active management while PVA > PVL (quiz C).

Defined benefit pensions: liabilities and a futures hedge ★★★★★

Liability model inputs: years of service G, wage W_0, wage growth w, multiplier m, years to retirement T, discount rate r (high-quality corporate), payout years Z

ABO (legal liability if plan closed now, lower) vs **PBO** (expected payout of ongoing plan, higher, usually more realistic)

Effective duration from PV_- and PV_+ with equal up/down curve shifts; equity duration assumed **zero**

```
PBO $2.57bn, effective duration 9.35 → BPV_L = 2.57bn ×
9.35 × 0.0001 = 2,402,950
Assets $3.07bn: 60% equity, 40% bond ladder (1–5 yr)
duration 2.85
BPV_A = 1.228bn × 2.85 × 0.0001 = 349,980 → gap 2,052,970
CTD BPV 128.98, duration 13.53, CF 0.9436 → contract BPV
136.69
100% hedge: buy 2,052,970 / 136.69 ≈ 15,019
Rates expected ↓ (110%): ≈ 16,521 ; rates expected ↑
(90%): ≈ 13,517
```

Daily margin: gains/losses on ~15,000 contracts paid in cash, while liability changes are unrealised → 100% hedges are rare; partial hedges common

One 1/32 move ≈ only 0.2 bp of yield on duration 13.53 → large margin flows

Closing the gap with interest rate swaps ★★★★★

In plain words: a receive-fixed swap acts like owning a fixed bond financed by floating debt → adds duration. Pay-fixed removes duration. Swaps are OTC, so traditionally no daily margin.

```
Swap BPV per 100 NP = (D_fixed - D_floating) × 100 × 0.0001
NP = BPV gap ÷ (swap BPV per 100 ÷ 100)
Pension: (9.18 - 0.25) → 0.0918 - 0.0025 = 0.0893 → NP = 2,052,970 / 0.000893 ≈ $2.30bn receive-fixed
Mandate 30–70%, 50% neutral
Expect rates ↑ → 30%: 0.3 × 2.30 = $0.69bn
Then expect rates ↓ → 70%: 0.7 × 2.30 = $1.61bn → add $0.92bn receive-fixed
Quiz: gap $300,000; (8.25 - 0.5) × 0.01 = 0.0775 → NP = 300,000 / 0.000775 = $387m
```

Many swaps are now marked to market and margined → less counterparty risk, but same cash-flow issues as futures

Quiz note: the quiz did not say which BPV is larger → size known, direction not

Swaptions for the duration gap (worked) ★★★★★

Swaption (named for the buyer's fixed leg)	Right to	Valuable when new swap fixed rate (SFR)...	Effect
Receiver swaption	Receive fixed at strike	Falls below strike	Adds duration when rates fall
Payer swaption	Pay fixed at strike	Rises above strike	Removes duration when rates rise

```
US plan: gap 450,000 (assets short); swap BPV 0.2571 per 100
1. 50% hedge: 450,000 × 0.5 / 0.002571 = $87.515m receive-fixed
2. Want 35% if rates rise → buy payer swaption on 15%: 450,000 × 0.15 / 0.002571 = $26.254m
   Premium 55 bp → 26.254m × 0.0055 = $144,397
3. Exercise if new SFR > 2.7% strike (pay 2.7% when market pays more)
```

Capture value: exercise and receive higher floating, or exercise and enter a new receive-fixed swap at the higher SFR and pocket the difference

Premium is a **sunk cost** — it does not affect the exercise decision

Swaption cost is limited to the premium; a swap can keep losing

Swaption collar: buy receiver + sell payer → cuts premium but gives away gains if rates rise above the payer strike

Choosing the optimal swap hedge (rate scenarios) ★★★★★

In plain words: all three hedges cut risk. Which is **best** depends on where you think swap rates will go. Plan needs more asset duration; must hedge.

Hedge	Optimal if new SFR is...	Why
Receive-fixed swap at 2.5%	**≤ 2.5%**	Like owning 2.5% bonds; gains as rates fall; no premium
Zero-cost collar: buy 2.3% receiver + sell 3.3% payer	**Between 2.5% and 3.3%**	Both options worthless → zero value; swap would lose, receiver cost premium
Buy 2.3% receiver swaption	**Well above 3.3%**	Loss capped at premium; swap and collar (short payer) keep losing

Breakeven above 3.3% where the receiver beats the collar is **not** covered

TRAP Seller of a payer swaption loses when rates rise above its strike.

Risks of liability-driven investing ★★★★★

Risk	Meaning
Convexity ignored	Duration-only hedges miss large-rate-move effects
Parallel-shift assumption	Duration assumes parallel moves
Structural risk	Curve twists; set asset convexity somewhat above liabilities but limit dispersion
Model risk	Many assumptions to model DB liabilities, effective duration, BPV
Measurement error	Weighted averages instead of cash-flow-based statistics
CTD can change	Futures duration and BPV change; accrued interest adjustment
Spread risk	Asset and liability rates differ (govt/swap vs corporate); Treasuries more volatile → non-parallel moves; swaps reflect MRR
Counterparty risk	OTC derivatives; collateral reduces it
Cash-flow (collateral) risk	Posting margin or collateral for futures and swaps
Asset liquidity risk	Cannot adjust positions quickly at fair cost

TRAP 'Liability rate = corporate, asset rate = government' → **SPREAD RISK** (quiz C). 'Cannot adjust at reasonable cost' → liquidity risk.

Bond indexing: why it is harder than equity ★★★★★

In plain words: bond index funds give cheap diversification and aim for minimal tracking error. **Pure indexing** (full replication) holds every bond; **enhanced indexing** matches the index's risk factors, not every holding.

Challenge	Detail
Size and variety	Far more issues, very different features → full replication impractical
Many issues per issuer	Different liquidity → concentrate in liquid issues
OTC dealer market	Post-2008 capital rules → smaller dealer inventories, wider spreads → less liquidity
Rare trading	Most bonds don't trade in a year; trades often unreported → **matrix / evaluated pricing** (infer yield from similar traded bonds); unusual features = hard
Index changes	New issues, maturities, rating changes alter composition quickly

Matching the primary risk factors ★★★★★

Match	Controls tracking error from
Modified duration (effective if options)	**Parallel** shifts
Key rate durations	**Non-parallel** shifts (quiz B)
Sector and quality weights; spread duration	Spread changes vs government bonds
Sector/coupon/maturity cell weights; callable and MBS weights	Convexity and option exposure (e.g. 1.7% in A-rated corporates of duration 2–3)
Issuer exposure	Issuer event risk (bankruptcy)

```
Effective duration = (PV₋ - PV₊) / (2 × Δcurve × PV₀)
Key rate duration = -(1/PV) × ΔPV / Δr_k
Spread duration: %Δ relative value ≈ -spread duration × Δspread
```

```
PV distribution of cash flows: 3-yr 4% semiannual bond at par (2% per period)
t: 1–5 coupons 2, t6 = 102 → PVs 1.961, 1.922, 1.885, 1.848, 1.811, 90.573 (sum 100)
w = PV/100 → 0.0196 … 0.9057 ; Σ t × w = 5.713 periods = 2.86 years
Each t × w = contribution to duration = a key rate duration
Match portfolio w to index w → matches KRDs, duration and convexity
```

Enhanced indexing: stratified sampling, ESG and adding value ★★★★☆

- **Stratified sampling (cell matching)**: split index into cells (e.g. 3 duration bands × 3 sectors = 9 cells); hold the same weight per cell (e.g. 5.1% in 1–5 duration Treasuries) with fewer bonds
- Can add **ESG** rules (exclude high-carbon issuers, include clean energy) while matching cell weights
- Add value or cut cost: lower expenses and trading costs · overweight cheap / underweight rich securities, sectors, curve points (avoid spread widening, favour narrowing) · overweight callables when rate volatility expected **low**, underweight when high

Passive bond exposure: other vehicles ★★★★★

Vehicle	Pros	Cons
Separately managed (full / enhanced)	Control, customisation	Higher setup cost
Index mutual fund (open-end)	Small investors; diversification; scale economies; buy/sell at NAV daily	Management fees, front/back loads; never matures; holdings and income change
ETF	Trades all day; **authorised participants** create/redeem in kind → arbitrage keeps price near NAV	Arbitrage weaker for illiquid bonds
Total return swap (TRS)	Receive index total return, pay MRR + spread = index bought with borrowed money; cheap entry	Not fully collateralised → leveraged; no ownership; **counterparty risk**; receiver pays negative index returns

- Indirect methods avoid big upfront trading costs but bring ongoing costs and other risks (quiz: TRS = A)

Selecting a bond benchmark ★★★★☆

- Start with client objectives and constraints → strategic asset allocation → index that matches (tactical discretion possible)
- Static bond portfolio: duration **falls** as bonds age (quiz: 'stays the same' is false)
- New issuance shifts index duration (issuers move shorter or longer) → index may stop fitting
- Custom index from desired duration and sectors, e.g. 50% 5–10-yr Treasuries + 50% 1–3-yr IG corporates
- **Smart beta / credit barbell**: long-term Treasuries supply duration (no spread risk); short-term corporates add spread return and suffer less if spreads widen

05 P5 Yield Curve Strategies

≈13% of PM marks · 16 subtopics · chapter priority ★★★★★

What this reading is about ★★★☆☆

In plain words: how to add value by positioning along the **government (credit-risk-free) yield curve**: what to do if you expect the curve to stay the same, or to change in level, slope or shape; how options and volatility fit; how key rate durations reveal a manager's view; how to invest across currencies; and how to estimate a strategy's return. (Credit comes in P6.)

LOS	Topic
P5.a	Factors affecting returns from benchmark yield changes
P5.b	Strategy when your view = market view (static curve)
P5.c	Strategy when your view differs: level, slope, curvature
P5.d	Strategy for expected volatility changes
P5.e	Key rate durations of portfolio vs benchmark
P5.f	Yield curve strategies across currencies
P5.g	Expected return and risks of a yield curve strategy

The yield curve and how it changes ★★★★★

In plain words: a yield curve plots yields (usually YTM) of one type of bond, normally government bonds, against maturity. Active management adds value only if your view is correct **and** not already in prices.

- Building a curve needs choices: interpolation to fill gaps, accounting/regulatory rules, which bonds to use
- Built from liquid, recently issued **on-the-run** bonds; using off-the-run bonds too would give a **higher** curve (liquidity premium)

Change	Meaning	Measure
Level	Parallel shift: all yields up or down equally — **explains most** bond price change	—
Slope (twist)	Curve steeper or flatter	Long yield − short yield (rising = steepening)
Curvature (butterfly)	Curve more or less bent	Butterfly spread = −short + 2 × medium − long (rising = more curvature)

Five-step return decomposition; convexity; bullet vs barbell ★★★★★

```
1. Coupon income = annual coupon ÷ current price
2. Rolldown = (price if curve unchanged - beginning
price) ÷ beginning price
3. Yield change = (-MD × ΔY) + (½ C × ΔY²)
4. Spread change = (-MD × ΔS) + (½ C × ΔS²)
5. Currency gain/loss (weighted by exposure)
```

Higher convexity → outperforms when rates are volatile → trades at a **premium** (lower YTM)

More cash-flow dispersion around duration → more convexity → **barbell** has more convexity than a **bullet** of the same duration

```
UK bonds: 2-yr MD 1.86, C 5.2 ; 10-yr MD 9.52, C 104.8 ;
20-yr MD 16.23, C 292.8
Bullet = 100% 10-yr ; barbell = 46.68% 2-yr + 53.32% 20-
yr (duration 0.4668×1.86 + 0.5332×16.23 = 9.52)
+50 bp: 2-yr -0.92% ; 10-yr -4.63% ; 20-yr -7.75%
Bullet -4.63% vs barbell 0.4668×(-0.92) + 0.5332×(-7.75)
= -4.56% → barbell wins
```

TRAP Barbell beats bullet for both **UP AND DOWN** parallel moves (quiz C).

Static yield curve: strategies when your view = the market's ★★★★★

In plain words: if you expect an upward-sloping curve to stay unchanged, earn more by taking more duration or leverage.

Strategy	How it earns	Main risk
Buy and hold	Duration longer than benchmark → higher coupons; low turnover and costs	Parallel shift **up** (quiz A)
Rolling down the curve	Coupon + **rolldown**: buy bonds at the end of a **steep** segment; yield falls as they age → price gain; sell and roll out	Curve shifts or flattens
Repo carry trade	Borrow short via repo, buy longer bonds (leverage)	Rates rise; funding cost
Long bond futures	Implicit leverage — only margin needed	Rates rise
Receive-fixed swap	Swap carry (fixed − MRR) + mark-to-market; fixed leg holds the duration	Rates rise

```
Futures BPV = BPV_CTD ÷ CF_CTD ; contracts to buy =
target BPV ÷ futures BPV
```

LIBOR replaced by **MRR** (market reference rates, often secured overnight)

TRAP Stable curve → **RECEIVE-FIXED** swap or **REPO** (borrowing). Pay-fixed or reverse repo (lending) are wrong (quiz A).

Worked: bond vs receive-fixed swap over six months ★★★★★

```
£50m par, 6-month horizon, rates fall 25 bp
Trade A: 10-yr 3% semiannual gilt, yield 2.5%, price
104.3998
Coupon = 0.03/2 × 50m = £750,000
New price: N 19, I/Y 1.125, PMT 1.5, FV 100 → 106.3828 →
gain (106.3828 - 104.3998)% × 50m = £991,500
At unchanged 2.5% yield: 104.2048 → rolldown = -£97,500 ;
yield change = +£1,089,000
Trade B: receive 3%, pay MRR 0.5%
Carry = (0.03 - 0.005)/2 × 50m = £625,000
Fixed leg at 2.75%: N 19, I/Y 1.375, PMT 0.015, FV 1 →
1.020776 × 50m = £51,038,817
Floating leg resets to par → MTM = +£1,038,817
```

Rolldown here is negative: the curve between 9.5 and 10 years is assumed flat, and the bond's premium pulls to par

Divergent view on level ★★★★★

In plain words: expect yields to fall (**bullish**) → lengthen duration. Expect yields to rise (**bearish**) → shorten duration.

Tool	Increase duration (bullish)	Decrease duration (bearish)
Cash bonds	Buy long / sell short bonds	Buy short / sell long bonds
Futures	Buy	Sell
Swaps	Receive fixed (pay floating)	Pay fixed (receive floating)

```
Benchmark equal-weight 2/10/20: D = (1.86 + 9.52 +
16.23)/3 = 9.203 ; C = 134.3
Bearish 50/25/25: D = 7.368 ; C = 102.0
+40 bp: benchmark -3.57% ; portfolio -2.87% → excess
+0.70%
Same with a swap on $100m: BPV target 73,680 - benchmark
92,030 = -18,350
Pay-fixed swap MD -8.354 → NP = -18,350 / (-8.354 ×
0.0001) = $21.98m
```

TRAP Expect rates down → **PAY-FLOATING** (= receive-fixed) swap (quiz A).

Divergent view on slope ★★★★★

In plain words: buy bonds whose yields will fall relative to the rest; short bonds whose yields will rise. Steepening → long short end, short long end. Flattening → long long end, short short end. Then set net duration by your level view: + if bullish, − if bearish, 0 if none.

View	What happens	Typical cause	Net duration
Bull steepener	Short rates fall more than long	Central bank cuts in a contraction	Positive
Bear steepener	Long rates rise more than short	CB too loose too long → inflation fears	Negative
Bear flattener	Short rates rise more than long	CB starts hiking	Negative
Bull flattener	Long rates fall more than short	Flight to quality	Positive

```
Steepener, no level view (duration-neutral): long £100m
2-yr → BPV = 100m × 1.86 × 0.0001 = £18,600
Short 10-yr: MV × 9.52 × 0.0001 = 18,600 → MV =
£19,537,815
2-yr -20 bp: +0.373% → +£373,040 ; 10-yr +20 bp: -1.883%
→ short gains ≈ £367,900
Total ≈ +£740,900
If next view is bull steepener → make duration positive:
buy more 2-yr or short less 10-yr
```

Don't memorise the table: buy where yields fall relative, sell where they rise relative, then set duration sign

TRAP Negative duration + long short end + short long end = **BEAR STEEPENER** (quiz B).

Divergent view on curvature: butterfly trades ★★★★★

In plain words: a butterfly combines a bullet (the **body**, medium maturity) with a barbell (the **wings**, short + long). Duration-neutral, but **not** convexity-neutral.

View	Middle yields	Butterfly spread	Trade
More curvature = **negative** butterfly twist	Rise vs wings	Rises	**Short body, long wings**
Less curvature = **positive** butterfly twist	Fall vs wings	Falls	**Long body, short wings**

```
Positive butterfly, duration-neutral, equal-value wings;
long £100m 10-yr → BPV £95,200
MV_w × 1.86 × 0.0001 + MV_w × 16.23 × 0.0001 = 95,200 →
MV_w = £52,625,760 short each wing
2- and 20-yr +10 bp, 10-yr -25 bp:
Body +2.413% → +£2.41m ; short 2-yr gains ≈ £0.10m ;
short 20-yr gains ≈ £0.85m → ≈ +£3.36m
```

Long body + short wings = **short convexity** → hurts if yields move equally; here the body moved more, so its convexity term was bigger

Memory aid: negative butterfly = the curve 'frowns' more

TRAP Long wings, short body = expects a **NEGATIVE BUTTERFLY TWIST** (quiz C) — the butterfly spread **RISES** .

Volatility views: embedded options ★★★★☆

Bond	Structure	Behaviour
Callable	Option-free bond + **short** call (issuer's)	Call gains as yields fall → price rises slowly → **negative convexity** at low yields
Putable	Option-free bond + **long** put (investor's)	Put gains as yields rise → price **floor** at high yields

$$\text{Effective duration} = \frac{PV_- - PV_+}{2 \times \Delta curve \times PV_0}$$

$$\text{Effective convexity} = \frac{PV_- + PV_+ - 2PV_0}{\Delta curve^2 \times PV_0}$$

Use **effective** measures — Macaulay/modified ignore uncertain cash flows

An embedded option always **lowers** duration vs the option-free bond

Expect rates ↑ → **least** likely to buy the **option-free** bond (falls most) — quiz C

Volatility views: stand-alone options ★★★★☆

Position	Gains when	Portfolio duration
Long call on bond or bond futures	Yields fall	Increases
Long put on bond or bond futures	Yields rise	Decreases
Long receiver swaption (right to receive fixed)	Rates fall	Increases
Long payer swaption (right to pay fixed)	Rates rise	Decreases

TRAP Expect yields down → **LONG CALL ON BOND FUTURES** (quiz C). Short receiver or long payer would lose.

Key rate durations: reading a manager's view ★★★★★

In plain words: a key rate (partial) duration shows the value change if **one** maturity's yield moves and others stay put. They sum to effective duration. Active KRD = portfolio KRD − benchmark KRD.

$$\text{KeyRateDur}_k = -\frac{1}{PV} \times \frac{\Delta PV}{\Delta r_k}$$

Zero-coupon bond: KRD = MD × weight (MD = maturity ÷ (1 + y)) — not for coupon bonds

Active KRD **> 0** at a maturity → expects that yield to **fall**; < 0 → expects it to rise

Worked example (3 zeros, equal-weight benchmark): active duration −3.875 → **bearish**; short end +0.98, long end 0 → short rates rise less → **steepener**; big negative middle, positive wings → **negative butterfly** → overall: bear steepener + negative butterfly

Quiz: active KRDs +3 (10-yr), −0.5 (20-yr), −3 (30-yr) via derivatives → **receive-fixed 10-yr, short 20-yr futures, pay-fixed 30-yr** (A)

Across currencies: unhedged foreign bonds ★★★★★

$$R_{DC} = (1 + R_{FC})(1 + R_{FX}) - 1$$

R_{FX} from the **direct** quote (DC/FC: domestic per 1 foreign)

Several markets: compute R_{DC} for each, then weight

```
UK investor, Japanese bond: JPY/GBP 150 → 125 ⇒ GBP/JPY
0.006667 → 0.0080 → R_FX = +20%
Australian buys US 2-yr zero at 96.374 → 99.939 after 1
yr: R_FC = 3.70%
USD/AUD 0.90 → 0.80 ⇒ AUD/USD 1.1111 → 1.2500 → R_FX =
12.5%
R_DC = 1.037 × 1.125 - 1 = 16.66%
```

TRAP Invert the quote if the domestic currency is the base. Direct quote up = foreign currency up = gain.

Hedging: covered vs uncovered interest parity ★★★★★

In plain words: hedging = sell the foreign currency forward. Under **covered interest rate parity (CIRP)** the high-rate currency trades at a forward **discount**, so a hedged foreign bond earns only your **domestic** risk-free rate.

$$F_{DC/FC} = S \times \left[\frac{(1 + r_{DC})}{(1 + r_{FC})}\right]^T$$

```
US 2-yr yield = (100/96.374)^½ - 1 = 1.864% ; AUD 2-yr 5%
F = 1.1111 × (1.05 / 1.01864)² = 1.1806 AUD/USD
Sell USD 100 forward → AUD 118.06 ; cost 96.374 / 0.90 =
AUD 107.0822
Return = (118.06 / 107.0822)^½ - 1 = 5.0% = AUD rate
USD at 3.1% forward premium: (1.1806/1.1111)^½ - 1
offsets lower USD yield
```

Uncovered parity (UIRP): high-rate currencies should actually depreciate to the forward rate → all unhedged investors earn the same

UIRP usually **fails**: spot ends between 1.1111 and 1.1806 → forward is a biased predictor = **forward rate bias** → carry trades profit on average

Quiz: EUR 1% (100/99.01 − 1), USD 3% → USD (high rate) at a 2% forward **discount**; 'sell USD at a 2% premium' is least likely (B)

Hedging foreign coupon bonds: fixed-fixed cross-currency swap ★★★★☆

```
AUD investor, USD 100m 10-yr Treasury at par, USD/AUD 0.80
Enter fixed-fixed cross-currency swap: pay USD fixed, receive AUD fixed; principals USD 100m / AUD 125m
Net: like owning an Australian bond
```

Building block	Legs
Swap A (USD)	Pay USD fixed, receive USD floating
Swap B (AUD)	Receive AUD fixed, pay AUD floating
Swap C (cross-currency basis)	Receive AUD floating, pay USD floating; exchange principals at start and end

Floating legs of A and B cancel with C; C funds the USD purchase and converts par back to AUD

Paying USD floating = borrowing USD → receive USD principal at start, repay at end

Negative basis: non-USD floating leg below CIRP fair rate → USD borrowers accept less on their own currency

Skip Swap A → net receive USD fixed, pay USD floating → profits if USD rates fall, currency still hedged

Unhedged cross-currency: the carry trade ★★★★★

In plain words: borrow in a low-rate currency, deposit in a high-rate currency. It earns the rate gap unless the high-rate currency falls.

```
Borrow USD 1-yr at 1% ; roll 90-day INR at 5% ; INR/USD 70
Stable: (1 + 0.05/4)⁴ - 1 = 5.1% → profit 4.1%
INR rates 5, 6, 4, 3%: (1.0125)(1.015)(1.01)(1.0075) - 1 = 4.6%
INR/USD 70 → 85: USD/INR 0.014286 → 0.011765 = -17.6%
Total = 4.6 - 1.0 - 17.6 = -14.0% (crisis / flight to quality)
```

Swap views across markets: receive floating where rates will rise, pay floating where they will fall; receive fixed where rates will fall, pay fixed where they will rise; watch FX

Quiz: foreign bull steepener + stable, higher-yield domestic curve → **pay foreign floating, receive domestic fixed** (C)

TRAP Carry trade profits mean **UNCOVERED** parity fails — covered parity is not involved (quiz B).

Evaluating a yield curve strategy (scenario analysis) ★★★★★

```
US manager, Canadian zeros, 1 year, CAD -0.5%, yields +25 bp
Coupon: 0 (zeros) ; spread: 0
Bullet rolldown (98.75 - 96.75)/96.75 = 2.067% ; yield -4.89×0.0025 + ½×28.5×0.0025² = -1.214%
Barbell rolldown (98.95 - 97)/97 = 2.010% ; yield -4.92×0.0025 + ½×43×0.0025² = -1.217%
USD: bullet (1 + 0.02067 - 0.01214)(0.995) - 1 = 0.35% ; barbell 0.29%
Bullet wins by 6 bp — sits on a steeper part of the curve (more rolldown)
```

```
Quiz: barbell 60% 2-yr + 40% 10-yr; bear flattener 2-yr +50, 10-yr +30
2-yr -0.92% ; 10-yr -9.52×0.003 + ½×104.8×0.003² = -2.81%
0.6 × (-0.92) + 0.4 × (-2.81) = -1.68%
```

TRAP Schweser recommends **COMPOUNDING** currency (multiplicative). Adding it, as in an earlier reading, is unlikely to be marked wrong.

06 P6 Fixed-Income Active Management: Credit Strategies

≈19% of PM marks · 12 subtopics · chapter priority ★★★★★

Credit risk, spreads and the credit cycle ★★★★★

In plain words: a credit spread is the extra yield a risky bond pays over a safe government bond. It pays you for two risks: the issuer may not pay (**credit risk**) and you may not be able to sell easily (**liquidity risk**).

```
Expected loss = POD × LGD ; LGD = 1 - recovery rate
CVA = PV of Σ(POD × LGD × expected exposure) → risky bond = risk-free bond - CVA
Simple fair spread ≈ POD × LGD
First lien spread 1.5%, recovery 50% → POD = 1.5%/0.5 = 3%
Second lien recovery 40% → spread = 3% × 0.6 = 1.8%
```

IG = BBB and above (main risk = **credit migration**/downgrade); HY = BB and below (main risk = default)

Credit spread curve = spread vs maturity; upward = defaults more likely later

Cycle: peak → spreads rising, curve upward sloping; recession → IG flattens, **HY inverts** (near-term default risk); HY moves more sharply

Rates and spreads move oppositely → HY bond price reacts less to rate changes: **empirical duration < analytical duration**

```
HY 6% vs Treasury 2% (spread 4%), MD 4.4: Treasury falls to 1%, spread widens to 5% → yield still 6% → price unchanged (empirical duration ≈ 0)
```

Fixed-rate spread measures ★★★★★

Measure	Definition	Note
Yield (benchmark) spread	Bond YTM − nearest on-the-run govt YTM	Simplest; maturity mismatch
G-spread	Bond YTM − **interpolated** govt YTM	Upward curve → g-spread < yield spread
I-spread	Bond YTM − interpolated **swap** rate	Tradeable hedge; carry
Asset swap spread (ASW)	Bond **coupon** − swap rate	Spread over MRR via bond + pay-fixed swap; bond below par → i-spread > ASW
Z-spread	Constant spread over the govt **spot** curve that prices the bond	Respects term structure
CDS basis	CDS spread − z-spread	Technical differences
OAS	Spread over an interest-rate tree after removing the option	**Only** valid measure for bonds with options — best for comparisons

```
Interpolation weight on longer bond = (M - S)/(L - S)
12-yr corporate 4.5% (coupon 4.3%, MD 11.1); govt 10-yr
3.0%, 20-yr 5.0%
Yield spread = 4.5 - 3.0 = 1.5% ; weight on 20-yr = 2/10
= 0.2 → interpolated 3.4% → g-spread = 1.1%
20-yr falls 10 bp → interpolated 3.38% → corporate yield
4.48% → price +11.1 × 0.02% = +0.222%
Swap rates 3.2% / 5.3% → interpolated 3.62% → i-spread =
0.88%, ASW = 4.30 - 3.62 = 0.68%
```

TRAP A bond whose OAS is very different from its other spreads has an embedded option — compare it using **OAS** .

Floating-rate note spreads ★★★★☆

```
Coupon = MRR + quoted margin (QM) ; required margin =
discount margin (DM)
DM > QM → below par ; DM < QM → above par ; DM = QM → par
1-yr FRN: MRR 0.55%, QM 1.25%, DM 1.55% → below par
Quarterly coupon (0.55 + 1.25)/4 = 0.45% → $4,500 ;
discount (0.55 + 1.55)/4 = 0.525%
Value of $1m ≈ $997,040
```

FRNs: low rate duration, but spread duration ≈ like a fixed bond (QM fixed)

DM assumes flat MRR; **Z-DM** uses forward MRR + spot discounting; upward MRR curve (recovery) → Z-DM < DM

Spread duration, DTS and excess spread ★★★★★

```
Spread return ≈ -EffSpreadDur × ΔSpread + ½ ×
EffSpreadCon × ΔSpread²   (Δspread = ΔOAS)
DTS = EffSpreadDur × spread
Portfolio: 70% A bond (9.0 × 180 = 1,620) + 30% C bond
(9.5 × 400 = 3,800) → DTS = 2,274
Index OAS 200 → +20 bp = +10% → portfolio falls 2,274 ×
0.1 = 227.4 bp
Expected excess spread = spread × t - EffSpreadDur ×
ΔSpread - POD × LGD × t
Spread 250 bp, dur 4.5, POD 1.5%, LGD 55%, spread → 270
in 6 months:
0.0125 - 4.5 × 0.0020 - 0.015 × 0.55 × 0.5 = -0.0625% →
overvalued
```

Spread changes are proportional to spread level → use DTS

Portfolio OAS is market-value weighted (include accrued interest)

TRAP De-annualise spread and POD × LGD for the holding period, and work in decimals.

Bottom-up credit: models & relative value ★★★★★

Model	How	Examples
Structural	Default when asset value falls below liabilities ('distance to default'); asset volatility from equity data	Moody's EDF, Bloomberg DRSK
Reduced form	Default intensity from macro + firm ratios	**Altman z-score**

```
z = 1.2 × WC/TA + 1.4 × RE/TA + 3.3 × EBIT/TA + 0.6 ×
MVE/TL + 1.0 × Sales/TA
> 3 low risk · 1.8–3 some risk · < 1.8 default likely
Example: 0.6, 0.54, 0.17, 2.5, 0.57 → z = 4.11 → low risk
```

```
New 10-yr issue: 5-yr spread 1.5 - 0.8 = 0.7% ; 15-yr
govt interpolated 0.75×2% + 0.25×4% = 2.5% → 15-yr spread
1.0%
10-yr spread = 0.5×0.7 + 0.5×1.0 + 0.05 premium = 0.90%
```

Bottom-up: universe → sectors → best relative value; operating history, profitability, leverage, coverage ratios

Lower equity volatility → farther from default → lower POD

Watch structure (options, seniority), liquidity, M&A

Top-down, factor and ESG credit strategies ★★★★☆

- Macro: strong growth → spreads narrow → move **down** in quality (more HY); weak growth → move up
- High spreads alone are not an opportunity — only if you foresee improvement before others
- Average rating: linear scale (1,2,3…) **understates** credit risk; non-linear (Moody's 1, 10, 20, 40…) is better; use DTS
- Credit factors: **carry** (OAS), **defensive** (low-risk/short duration earns better risk-adjusted), **momentum**, **value**
- ESG: negative screening · ESG ratings (correlated with credit ratings) · direct funding e.g. **green bonds**

Liquidity risk & tail risk (VaR) ★★★★★

On-the-run sovereigns very liquid; small off-the-run corporates may not trade → matrix pricing; TRACE for US corporate trade data

Manage liquidity: liquid bonds for short-term strategies · CDS and ETFs (APs create/redeem) · **pay-fixed asset swap** while slowly selling a big illiquid bond, unwinding as you sell

VaR method	Pro	Con
Parametric (normal; 5% = 1.65σ, 1% = 2.33σ)	Simple	Poor for non-normal (options)
Historical simulation	Real data, non-normal OK	Depends on the sample
Monte Carlo	Flexible	Model risk

```
$100m bond, MD 12, annual yield vol 1.5% → monthly 1.5%/
√12 = 0.433%
1% VaR yield move = 2.33 × 0.433% = 100.89 bp → loss 12 ×
1.0089% = 12.107% → $12.1m
```

VaR flaws: tails worse than forecast, ignores correlation/liquidity shifts, no tail-size info

CVaR = average loss in the tail · **IVaR** = VaR change from adding/removing a position · **Relative VaR** vs benchmark

Stress the consequences: margin calls, sponsor contributions, bank reserves

Credit default swaps: pricing ★★★★★

In plain words: a CDS is insurance on a bond. The **buyer** of protection pays a fee and gets paid if the issuer defaults — he is 'short risk', like being short the bond. The **seller** is 'long risk', like owning the bond.

```
Standard coupon: 1% IG, 5% HY
Upfront premium (paid by buyer) ≈ (CDS spread - fixed coupon) × EffSpreadDur
CDS price ≈ 1 + (fixed coupon - CDS spread) × EffSpreadDur
HY, spread 5.5%, dur 4.5 → upfront 2.25% ; price 97.75
Spread falls to 5.4% → +0.45% price → buyer loses 0.45%
Quiz: HY 4.7%, dur 4.5, $10m → seller pays buyer (0.05 - 0.047) × 4.5 × 10m = $135,000
```

Single-name or index (CDX NA IG = 125 names); payer CDS option = right to buy protection (profits if spreads above strike); receiver = right to sell protection

Long risk position behaves like a long bond: profits when spreads fall and CDS price rises

CDS long/short and curve trades ★★★★★

```
Long XYZ / short ABC: sell XYZ protection, buy ABC protection, $15m each
ABC +15 bp: 15m × 0.0015 × 4.59 = $103,275 ; XYZ -20 bp: 15m × 0.002 × 4.62 = $138,600 → $241,875
Flattener (HY index): buy 5-yr protection $5m (dur 4.78 → BPV $2,390), sell 10-yr protection $2,390/(9.18×0.0001) = $2,603,486
5-yr -5 bp → -$11,950 ; 10-yr -20 bp → +$47,800 → +$35,850
```

Buy protection where spreads will widen; sell where they will narrow

Curve trade: buy protection at maturities expected to rise relative to others

Static & dynamic credit curve strategies ★★★★★

View	Cash bonds	CDS
Static curve	Lower average rating or extend spread duration (buy and hold longer)	Sell protection on lower quality or longer tenors
Economic slowdown (HY widens more; curves flatten/invert)	Buy IG, sell HY (flight to quality)	Buy HY protection, sell IG; buy short-dated, sell long-dated protection
Economic recovery (HY narrows more; curves steepen)	Buy HY, sell IG	Sell HY protection, buy IG; sell short-dated, buy long-dated protection

```
Static, cash: overweight $10m 15-yr corporate vs 10-yr for 6 months
Extra coupon $35,000 (govt part $15,000, credit $20,000)
Rolldown: 10-yr → 100.8888 to 101.1053 = $21,650 ; 15-yr → 102.4187 to 103.0726 = $65,390 → extra $43,740
Total $78,740; from credit = $20,000 + $32,695 = $52,695
```

```
Static, CDS: sell $50m 10-yr HY protection; 5-yr 450 bp, 10-yr 550 bp → 9-yr 530 bp
Price 0.9555 → 1 + (0.05 - 0.053) × 8.1 = 0.9757 → +$1,010,000 + coupon $2,500,000 = $3.51m
Slowdown: sell IG protection + buy HY protection, $50m each, spreads +40% / +80%
Net coupon -$2m ; HY leg +$4,832,500 ; IG leg -$770,000 → +$2,062,500
```

```
Quiz: sell IG protection $10m, spread 150 → 120 bp, dur 4.5 → 3.5
Price 0.9775 → 0.993 = $155,000 + coupon $100,000 = $255,000
```

Global credit, emerging markets ★★★★☆

One-industry/commodity EM economies give little sector diversification

Sector mix differs (US has far more MBS/ABS)

Accounting differences distort ratios

Credit cycles differ in timing → opportunities

```
Recovery view, buy European HY credit via CDS (sell protection): spread 4.5%, dur 4.55
Price 1 + (0.05 - 0.045) × 4.55 = 1.02275 → spreads -20% → 1 + (0.05 - 0.036) × 4.55 = 1.0637 → +4%
EUR +2% → 1.04 × 1.02 - 1 = 6.08%
```

Sovereign analysis: institutions (property rights, politics, geopolitics, ESG) · economy (tax base, debt/GDP, deficit/GDP — fiscal stability) · FX regime (external debt/GDP = distress risk; reserves/GDP = liquidity)

EM corporates: government influence, **sovereign ceiling**, liquidity premium, currency risk (UIRP often fails)

Structured credit & fixed-income analytics ★★★★☆

Reason to use	Instruments
Tranching creates new risk profiles (losses hit junior first)	CDOs (general debt), CLOs (leveraged loans)
Access collateral you can't buy directly	MBS, ABS (cards, autos), **covered bonds** (on bank's balance sheet, recourse to the bank, bad loans substituted → lower risk)

Recovery + housing rebound → sell AA MBS tranche, buy BB MBS tranche

Analytics = inputs (positions, market data, credit/ESG ratings, index data) + user parameters (term-structure, VaR, scenario models, filters, horizon, objective) + outputs (exposures vs benchmark, construction, risk/scenario, trading & cash tools)

MBS hold a short prepayment option → **lower** volatility assumption raises their value

07 P7 Trade Strategy and Execution

≈12% of PM marks · 8 subtopics · chapter priority ★★★★☆

Why managers trade ★★★★★

Motivation	Key points
Profit seeking	**Alpha decay** → urgency (news traders high, fundamental low); avoid **information leakage** using several venues incl. **dark pools** (ATS in US; MTF / systematic internaliser in Europe) vs **lit** exchanges
Risk management / hedging	Rebalance to target exposure (duration, FX hedge); derivatives if liquid and allowed; watch leverage
Cash flows	Subscriptions/redemptions; **cash drag** → equitise with ETFs/futures; redemptions at NAV → trade at **close**; weigh liquidity and tax when choosing what to sell
Corporate actions, margin calls, index reconstitution	Reinvest income; urgent sales for margin calls; reconstitution → trade at close to cut tracking error

Inputs to trading strategy & the trader's dilemma ★★★★★

Input	Effect
Order: side, absolute size, size vs ADV	Buying into rising market costlier; one-sided lists carry more market risk; bigger orders → more impact
Security: type, short-term alpha, volatility, liquidity	Fast alpha decay → urgent; high volatility → **execution risk**; liquidity (tight spreads, depth) lowers both costs
Market conditions	Crises: volatility up, liquidity down (2008; quant crowding Aug 2007)
Risk aversion	More averse → trades faster

Trader's dilemma: trade fast → market impact & leakage; trade slow → execution risk

Longer horizon lowers market impact but raises execution risk

Higher liquidity lowers **both**

Reference price benchmarks ★★★★★

Type	Benchmark	Used by
Pre-trade	Decision price	Ideal measure of the PM's decision
	Previous close	Quant managers (proxy for decision)
	Opening price	Fundamental long-term managers (ignores overnight news); not if trading in the opening auction
	Arrival price (when order reaches market)	Alpha-seeking active managers
Intraday	VWAP	Passive over the day, rebalancing (sells fund buys)
	TWAP	Ignore volume outliers; erratic volume
Post-trade	Closing price	Index funds, NAV-based redemptions; can't judge during trading
Price target	Manager's fair value	Short-term alpha: buy all you can below target

Short-term alpha trade: high urgency, price target + arrival price, algorithm

Long-term alpha in illiquid bonds: low urgency, sell gradually over weeks

Index reconstitution, low TE mandate → **market on close**

Execution methods & algorithms ★★★★★

Method	Description
High-touch principal (broker risk)	Dealer takes risk, priced into spread; OTC, RFQ markets; very large/illiquid/urgent
High-touch agency	Broker finds the other side; risk stays with manager
Electronic / DMA	Order-driven limit order book; buy side accesses exchange through broker's technology
Profit-seeking algos	Market makers, quants, HFT

Execution algorithm	Best for
Scheduled: POV (participation), VWAP, TWAP	Small orders, liquid markets, low urgency, minimise impact (rebalancing)
Liquidity-seeking (opportunistic)	Large orders, illiquid, higher urgency, fear of leakage, sporadic liquidity; lit + dark
Arrival price	Small orders, liquid; prices expected to move against you; risk-averse to execution risk
Dark strategies / liquidity aggregators	Large orders, illiquid, need not fill immediately
Smart order router (SOR)	Small market or limit orders across many venues

POV: uses liquidity when it appears but trades at any price and may not finish

VWAP/TWAP: finish on time but may force trades in thin periods

100% of ADV → high-touch principal; big + urgent + illiquid → liquidity-seeking; small, low urgency → scheduled

New: **clustering** (ML groups trades to pick algos) · high-frequency forecasting with **LASSO** to cut variables

Trading by market ★★★★☆

Market	How it trades
Equities	Lit exchanges + dark pools; electronic; algos common
Fixed income	Many heterogeneous issues, low liquidity, big tickets → dealer quote-driven, high-touch, electronic **RFQ**; algos only for on-the-run Treasuries & futures; urgent → principal, else agency
Exchange-traded derivatives	Electronic, DMA; algos less common (more futures than options)
OTC derivatives	Dealer, high-touch; post-2008 central clearing push (Dodd-Frank 2010)
Spot FX	OTC tiers: interbank, interdealer, bank-to-client; large urgent → RFQ; large non-urgent → scheduled algos or agency; small → DMA

Example: small urgent futures → DMA · agency MBS → high-touch broker · large non-urgent equity → scheduled algorithm

Implementation shortfall (worked) ★★★★★

```
IS = paper return - actual return  (paper = all shares at
decision price, no costs)
Buy 50,000 SJB, decision $20.00, arrival $20.10, filled
40,000 at $20.34, commission $0.02, close $20.55
Paper = 50,000 × 0.55 = 27,500 ; actual = 40,000 × 0.21 -
800 = 7,600 → IS = $19,900 = 199 bp
Delay = 40,000 × (20.10 - 20.00) = 4,000 (40 bp)
Trading = 40,000 × (20.34 - 20.10) = 9,600 (96 bp)
Opportunity = 10,000 × (20.55 - 20.00) = 5,500 (55 bp)
Fixed fees = 800 (8 bp)
```

```
Sell 100,000 FTRB, decision £2.56, arrival £2.59, sold
70,000 at £2.60, fee £400, close £2.54
Paper = 100,000 × 0.02 = 2,000 ; actual = 70,000 × 0.06 -
400 = 3,800 → IS = -£1,800 = -70.3 bp (benefit)
Delay -82.0 · trading -27.3 · opportunity +23.4 · fees
+15.6
Quiz: sell 10,000 at $32.50, 4,000 filled at $32.65,
$0.03/sh, close $32.35 → paper 1,500, actual 1,080 →
12.92 bp
```

Delay cost ↓ with faster pre-trade analysis

Opportunity-cost analysis → redeploy unfilled cash into the next idea

Negative cost = trader added value

TRAP For sells, flip the signs: a higher sale price is a gain.

Evaluating execution: arrival, market-adjusted, added value ★★★★★

```
Cost (bp) = Side × (execution price - benchmark) /
benchmark × 10,000   (Side +1 buy, -1 sell)
Index cost (bp) = Side × (index VWAP - index arrival) /
index arrival
Market-adjusted cost = arrival cost - β × index cost
Added value = arrival cost - estimated pre-trade cost
Buy at $25.50 vs close $25.60 → -39.1 bp (benefit)
Buy: arrival €10 → €10.15 = 150 bp ; index 3,500 → 3,507
= 20 bp ; β 1.5 → 150 - 30 = 120 bp
Pre-trade estimate 160 → added value -10 bp (good)
Quiz sell: 8.40 → 8.38 = 23.8 bp ; index cost -1 ×
50/3000 = -166.7 ; β 1.3 → 240.5 bp
```

Market-adjusted cost stops traders being blamed or credited for market moves

Trade governance & best execution ★★★★★

Trade policy area	Contents
1. Meaning of best execution	Best **result**, not just best price: price, costs, speed, likelihood of execution & settlement, size, nature of trade (block sale at a discount can be best execution)
2. Factors for optimal approach	Urgency, size, liquidity, security type, venues, strategy horizon, reason; regulations — **MiFID II** effectively bans soft dollars in Europe
3. Approved brokers & venues	Kept by a **Best Execution Monitoring Committee** (execution, compliance, risk); criteria: execution quality, financial stability, ethics, settlement, costs, willingness to commit capital
4. Monitoring	Reputation, errors, crimes, solvency; remove failing brokers; keep records for clients and regulators

TRAP 'Use the lowest-cost broker' and 'traders alone decide governance' are both **NOT** best practice.

08 P8 Case Study: Institutional (University Endowment)

≈15% of PM marks · 10 subtopics · chapter priority ★★★★★

How to read this case ★★★★★

In plain words: this reading has almost no new theory. It takes one university endowment (QU) and asks you to apply ideas from the whole curriculum: liquidity, asset allocation, ethics, derivatives, rebalancing and ESG. Expect it as an **essay set**: 'justify', 'recommend', 'identify the violation'.

Answer with the case numbers: quote the % or bp from the exhibit, then give the reason

One recommendation + one reason per point earns the marks; long lists do not

Managing liquidity risk: four tools ★★★★★

Tool	What it does
1. Liquidity profile & time-to-cash table	Sorts assets by how fast they turn into cash (e.g. highly liquid, liquid, semi-liquid, illiquid). **Liquidity budget**: minimums for liquid tiers, a **maximum** for the illiquid tier. Exit > 1 year = illiquid. Same asset can differ by vehicle (ETF vs commingled fund)
2. Rebalancing & commitment policy	**Systematic rebalancing**: calendar or percent-range (wider ranges for volatile assets). **Automatic adjustment**: PE 2% under target (β ≈ 1) → hold +2% public equity → same systematic risk, less illiquidity. **Commitment pacing** model (drawdown and distribution rates); diversify across funds and vintages; scenario analysis
3. Stress testing	Test assets **and** liabilities together: high volatility, low returns, high correlation, low liquidity, higher payouts. Base on history, statistical models or hypothetical scenarios
4. Derivatives	Little cash needed. Futures overlay rebalances without touching external managers. Buy options = cheap leverage; sell options = premium income (liquidity)

TRAP Calendar and percent-range rules are **SYSTEMATIC** rebalancing policies — a quiz statement that calls them something else is wrong.

The illiquidity premium ★★★★☆

In plain words: investors want extra return for money they cannot get back quickly. The longer the lock-up, the bigger the premium.

```
Option view: price of illiquid asset = price of same
marketable asset - value of a put (the right to sell you
gave up)
Illiquidity premium = expected return of illiquid asset -
expected return of marketable asset
```

Hard to measure: many other factors drive returns

Estimates use broad indexes, but a real investor holds few funds → more **unsystematic** risk → needs a liquidity budget

TRAP 'The illiquidity premium is easy to measure' is FALSE .

QU endowment: facts you need ★★★★★

Item	Detail
Size	$8bn: 75% unrestricted, 25% donor-restricted
Role	Pays about 60% of the $583m operating budget
Spending rule	Changed from 5% of beginning value to a **geometric smoothing** rule (uses prior spending $358.1m and prior year-end value $7,002.3m)
Return need	7–8% nominal = 5% spending + 2–3% inflation
Risk target	Standard deviation 12–14%
Manager	QUINCO: 11-person board, 13 staff, **external managers only**, 5 asset classes; Board approves managers; CIO Winter leads the investment committee

Asset class	1996	2017
Cash	1%	1%
Domestic stocks + bonds	~70%	~30%
International developed equity	25%	10%
Emerging markets	0	10–15%
Private equity	< 5%	15–20%
Real assets	< 5%	10–15%
Diversifying strategies	0	10–15%

Review: PE returns are smoothed → **unsmooth** them and volatility rises

QU lagged peers because it took less risk and held fewer illiquid assets

Proposed allocation: justify it & name the trade-offs ★★★★★

In plain words: CIO Thompson wants more private equity and real assets. You must argue for it, name what it costs, and say what to check.

Metric	Existing	Proposed
Sharpe ratio	0.33	0.34
Standard deviation	12.5%	13.2% (+0.7%)
P(keep ≥ 75% purchasing power, 20 yrs)	65%	70%

Why more illiquids: long horizon lets QU earn the illiquidity premium → efficient frontier shifts up · 20 years of good private-asset results with expert managers · QU is under-weight illiquids vs peers

Thompson's reasons: lower expected returns everywhere · QU takes less risk than peers · fixed income has a low Sharpe ratio · Monte Carlo shows better purchasing power

Trade-offs: higher volatility (but needed to earn the same return) · higher fees (exhibit returns are already **net of fees**)

Reduce the new risk: many smaller commitments (less unsystematic risk) or cash-flow modelling under stress

Before approving, confirm: 13.2% SD and 30% chance of 25% erosion fit risk appetite · new mix still meets all liquidity needs

TRAP More illiquids raise UNSYSTEMATIC risk — diversify across many funds and vintages.

Why liquidity gets worse in a crisis ★★★★★

Cause	Effect
Capital calls > distributions	PE funds call cash (a legal duty) while exits stop
Gates / lock-ups	Hedge funds may block withdrawals
Lagged PE valuations (denominator effect)	Public assets fall now, PE marks fall later → PE share rises, liquid share falls
Spending rule is countercyclical	Smoothed spending is > 5% of value after a fall → more cash needed exactly when markets are weak

Profile (highly liquid / liquid / semi / illiquid)	Normal	Stress
Existing	19 / 26 / 22 / 33	15 / 26 / 20 / 39
Proposed	14 / 24 / 23 / 39	11 / 25 / 21 / 43

Proposed: highly liquid −5% and illiquid +6% (normal); −4% and +4% (stress)

Follow-up: cash-flow forecasts, commitment pacing, liquidity budget, regular stress tests (historical + hypothetical), monitoring at high-risk times

Ethics in manager selection (who broke what) ★★★★★

In plain words: a donor-trustee's firm (GVC) is a finalist for a QU mandate. Many people act badly. For each person, name the Standard and the fact.

Person	Standards	Why
Aaron Winter, CIO	VI(A), I(B), III(E), I(C)/III(D), V(A)	Did not disclose GVC owner's link to QU or that GVC's MD is a former colleague; under pressure; suspected confidential data and inflated returns but did not report or investigate
Virginia Hall, trustee & GVC owner	VI(A), IV(A), III(E)	Conflict (trustee + finalist owner); put own business before QU; may have obtained confidential data
QU president & treasurer	IV(A), III(E), VI(A)/I(B)	Pressured Winter to hire GVC; may have leaked data; should disclose bias and abstain from the vote
Jason Allen, GVC MD	I(C), III(D)	Presented possibly wrong / inflated performance
Bud Davis, senior PM	VI(A), I(B)	Spouse is Hall's daughter; recommended GVC without disclosing

Standards list: I(B) Independence & Objectivity · I(C) Misrepresentation · III(D) Performance Presentation · III(E) Confidentiality · IV(A) Loyalty · V(A) Diligence · VI(A) Disclosure of Conflicts

GVC context: raising $300m Fund II after a $100m Fund I; it cut its fee; rival finalist Beacher

TRAP Non-CFA people (president, treasurer) are judged by the same tests in this case answer — apply the Standards to each actor named.

Equity exposure: ETF vs futures vs total return swap ★★★★★

In plain words: QU wants +1% US large-cap ($80m) for tactical asset allocation. Compare cost, work, tracking error and risk.

	ETF	Futures	Total return swap
Cost	**35 bp** (70% impact + fee)	42 bp (half is roll cost)	51 bp
Logistics	None (ETF manager runs it)	Quarterly roll, daily margin	None
Tracking error	Cash drag, diversification rules	Liquidity, rate differences	**None** (receives index return)
Rate / counterparty risk	None	Interest rate risk	Rate + counterparty
Overall (no leverage)	**Best**		

```
With leverage 4 (25% cash, 75% financed): financing 2%
for futures/swaps, 2.5% for ETF
ETF: 80m × 75% × 2.5% = $1.5m = 1.875% → total 0.35 +
1.875 = 2.225%
Futures: 80m × 75% × 2% = $1.2m = 1.5% → total 0.42 + 1.5
= 1.92% (cheapest by 9 bp)
Swap: 1.5% + 0.51 = 2.01%
```

Policy changes: tracking error limit 100 → 250 bp (150 bp for TAA); max leverage 5%; TAA uses fair-value / mean-reversion models (1–3 yrs, backtested)

ETF on 50% margin: $80m exposure = $40m cash + $40m borrowed

Rebalancing: cash market or derivatives? ★★★★★

Policy: rebalance quarterly; check drift monthly; liquid classes back to target or corridor edge; illiquid classes via new commitments

Now: int'l developed equity 11.5% (top of range 11%) · fixed income 6.5% (target 9%, range 6–12%) · PE 19.2% (low end 18%) · real assets 13.8% (low end 13%)

Trade	Cash market	Futures	Choose
0.5%, ~3 months	30 bp (spread 5 + impact 5 + cash drag 20)	24 bp (3 + 4 + tracking 17)	**Futures**
2.5% (to 9% target), ~1 year	60 bp (5 + 5 + cash drag 50)	82 bp (4 + 4 + tracking 68 + roll 6)	**Cash** (saves 22 bp)

Cash market: cash drag (delays, disrupting active managers) = two-thirds of cost

Futures: short equity + long bond futures → two contracts → tracking error 17 bp vs 8 bp for one; tracking error > 70% of cost

Rule: small, short-term, reversible, needs speed → derivatives overlay (no effect on active managers). Large, long-term, permanent → cash market

TRAP Derivatives' three roles here: **REBALANCING, TAA CHANGES, SHORT-TERM LIQUIDITY** .

ESG policy & the child-labour case ★★★★★

In plain words: a portfolio company (Portro) used a supplier with child labour (ILO report). Students demand divestment. What should QU do?

QUINCO ESG policy: ESG and financial factors both affect risk and return · transparency · long-term sustainability

Manager due diligence (added by Natalya Long, CFA): formal ESG policy? · how ESG enters the process · timely disclosure of material ESG issues · fit with QUINCO policy

Responsible ownership: proxy voting and engagement (human capital, climate, disclosure) · reporting · communication · UN PRI signatory

Monitoring metrics: ESG scores vs peers and policy benchmark · sensitivity to climate risks · portfolio carbon footprint · track trends over time

Response: engage the manager and Portro · join other investors · vote proxies · remove the supplier's goods from the campus store · divest only as a **last resort**

TRAP Divestment is **NOT** the first answer — engagement is preferred.

F Formula Sheet (all readings)

≈revise last marks · 4 subtopics · chapter priority ★★★★★

Equity (P1–P3) ★★★★★

```
Tracking error = SD(Rp - Rβ)
E(R_A) = IC × √BR × σ_A × TC
Active Share = ½ Σ |wpi - wβi|
Contribution to variance CVi = wi × Cov(ri, rp) ; % = CVi / σp²
Contribution to active variance = (wpi - wβi) × Σj (wpj - wβj) Cov(RAi, RAj)
Gross exposure = long + |short| ; net = long - short
R_g ≈ R_a - σ²/2
```

Liability-driven & yield curve (P4–P5) ★★★★★

```
MD = Macaulay duration / (1 + periodic YTM)
%ΔValue ≈ -MD × ΔY + ½ C (ΔY)²
BPV = MD × value × 0.0001
Futures BPV ≈ BPV_CTD / CF_CTD ; contracts = (BPV_L - BPV_A) / BPV_futures
Swap NP = BPV gap / swap BPV per 1 of notional ; swap BPV = (D_fixed - D_float) × 0.0001
Effective duration = (PV- - PV+) / (2 × Δcurve × PV0)
Effective convexity = (PV- + PV+ - 2PV0) / (Δcurve² × PV0)
Key rate duration = -(ΔPV / PV) / Δkey rate
Butterfly spread = -short + 2 × medium - long
R_DC = (1 + R_FC)(1 + R_FX) - 1
Forward (CIRP) F = S × [(1 + r_DC)/(1 + r_FC)]^T
```

Bonds with options: use effective, not modified, duration

Credit (P6) ★★★★★

```
Credit spread ≈ POD × LGD
EffSpreadDur = (PV- - PV+) / (2 × Δspread × PV0)
EffSpreadCon = (PV- + PV+ - 2PV0) / (Δspread² × PV0)
%Δprice ≈ -EffSpreadDur × Δs + ½ × EffSpreadCon × (Δs)²
%Δrelative value ≈ -spread duration × Δspread
DTS = EffSpreadDur × spread
Expected excess spread ≈ spread - (EffSpreadDur × Δspread) - (POD × LGD)
FRN discount margin: price from (MRR + DM) vs coupon MRR + QM
CDS price ≈ (fixed coupon - CDS spread) × EffSpreadDur_CDS
```

Trading (P7) ★★★★★

```
IS = paper return - actual return
= delay + trading (execution) + opportunity cost + fixed fees
Cost (bp) = Side × (P_exec - P_bench)/P_bench × 10,000 (buy +1, sell -1)
Market-adjusted cost = arrival cost - β × index cost
Added value = arrival cost - pre-trade estimated cost (negative = trader added value)
```